MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE RHF

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.043 failures per million operating hours for class I or 0.097 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be RHF.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/297-1 for class I and M28787/297-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

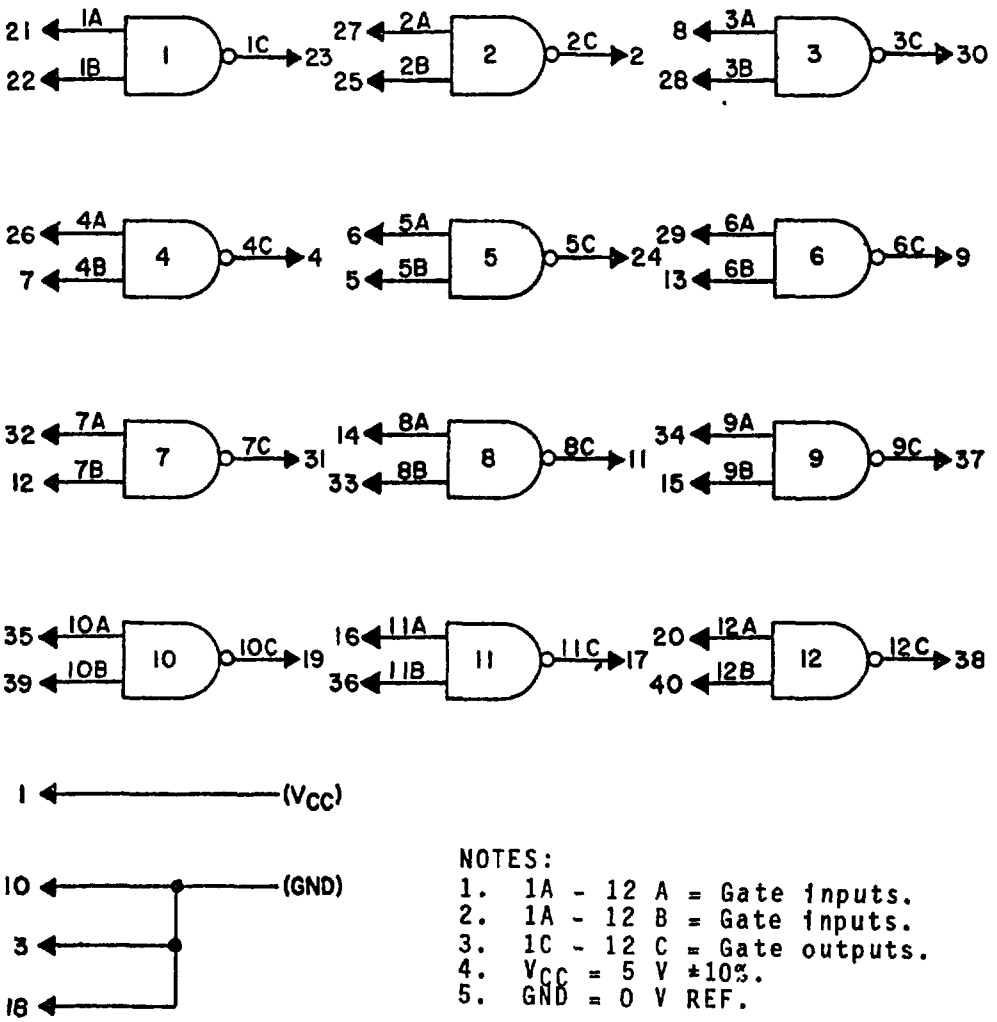
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. 1A - 12 A = Gate inputs.
2. 1A - 12 B = Gate inputs.
3. 1C - 12 C = Gate outputs.
4. V_{CC} = 5 V ±10%.
5. GND = 0 V REF.

*FIGURE 1. RHF module function diagram, NAND gate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B12	V_{OUT} (low loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C24	Input breakdown current	V	S	0	100	0	100	0	100	0	100	0	100	µA
D1-D12	Output leakage current	VI	X	0	1	0	1	0	1	0	1	0	1	mA
E1-E12	Output short circuit current	VII	X	-15	-100	-15	-100	-15	-100	-15	-100	-15	-100	mA
F1-F24	Input clamp diode voltage	VIII	X	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G12	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H24	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J24	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
K1-K24	Input current, low level	XII	X	-10	-430	-10	-430	-10	-430	-10	-430	-10	-430	µA
L1,L2	Ground pin continuity	XIII	X	0	1	0	1	0	1	0	1	0	1	Ω
L3,L4	Ground pin continuity	XIII	Q	0.00	0.10	0.00	0.10	0.00	0.10	0.00	0.10	0.00	0.10	Ω
M1	Power supply current, low level outputs	XIV	S	0.1	17.6	0.1	17.6	0.1	17.6	0.1	17.6	0.1	17.6	mA
M2	Power supply current, high level outputs	XIV	S	0.1	6.4	0.1	6.4	0.1	6.4	0.1	6.4	0.1	6.4	mA
N1-N12	Transition time (t_{TLH}), C_L = 50 pF	XV	S	5	50	5	50	5	55	5	50	5	55	ns
N13-N24	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	10	80	10	80	10	90	10	80	10	96	ns
P1-P12	Transition time (t_{THL}), C_L = 50 pF	XV	S	3	18	3	18	3	20	3	18	3	20	ns
P13-P24	Transition time (t_{THL}), C_L = 150 pF	XV	Q	6	40	6	40	6	42	6	40	6	42	ns

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q1-Q12	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	3	19	3	19	3	21	3	19	3	21	ns
Q13-Q24	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	4	29	4	29	4	31	4	29	4	31	ns
R1-R12	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	3	19	3	19	3	21	3	19	3	21	ns
R13-R24	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	4	29	4	29	4	31	4	29	4	31	ns
S1-S12	Output crosstalk, low level	XVI	Q	0	500	0	500	0	500	0	500	0	500	mV
T1-T12	Output crosstalk, high level	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
U	Power supply transient amplitude	XVII	Q	0	400	0	400	0	400	0	400	0	400	mV
V	Power supply filter capacitance	XVIII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF
W1-W24	Input capacitance	XIX	Q	5	15	5	15	5	15	5	15	5	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through W shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through W in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through T and V shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition V, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through T and V shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through W shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits from table I.

	Class I	Class II	Unit
Maximum power dissipation			
All outputs at low level (outputs not loaded)	97	97	mW
All outputs at high level (outputs not loaded)	35	35	mW

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.

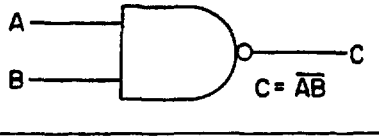
	Class I	Class II	Unit
Input voltage			
Low level	-0.5 V min, 0.7 V max	-0.5 V min, 0.7 V max	V
High level	2 V min, V_{CC} max	2 V min, V_{CC} max	V
Input current			
Low level (V_{IL} = 0 V)	-430 max	-430 max	µA
High level (V_{IH} = 2.5 V)	20 max	20 max	µA
Output voltage			
Low level	400 max	400 max	mV
High level	2.5 min	2.5 min	V
Output current			
Low level (V_{OL} = 0.4 V)	4 max	4 max	mA
High level (V_{OH} = 2.5 V)	-400 max	-400 max	µA
Propagation delay time			
t_{PHL} (C_L = 50 pF)	3 min, 21 max	3 min, 21 max	ns
(C_L = 150 pF)	4 min, 31 max	4 min, 31 max	ns
t_{PLH} (C_L = 50 pF)	3 min, 21 max	3 min, 21 max	ns
(C_L = 150 pF)	4 min, 31 max	4 min, 31 max	ns

* 6.5 Module function. This module contains 12 independent 2-input NAND gates.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship.

Input		Output
A	B	C
L	L	H
L	H	H
H	L	H
H	H	L



* 6.5.2 Parallel outputs. Paralleling of outputs to obtain extended drive is permissible only when parallel inputs and outputs are confined within each group specified below. The output drive capability of this combination is equal to the sum of the output drive capabilities of the individual circuits. The circuit groups which can be paralleled are:

a. Circuits 1, 2, 3 and 4.
b. Circuits 5, 6, 7 and 8.
c. Circuits 9, 10, 11 and 12.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.

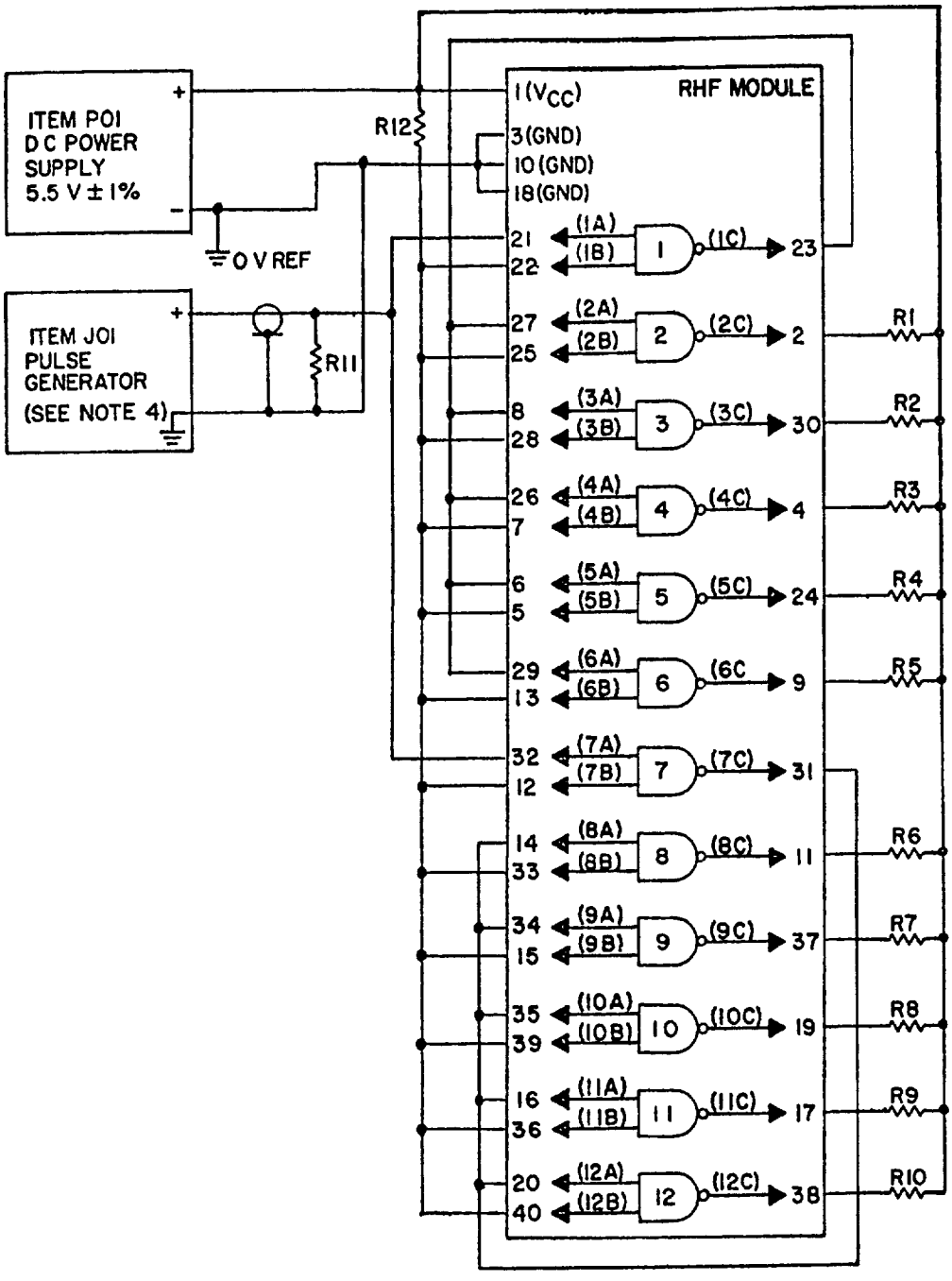
6.6 Changes for previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N476)

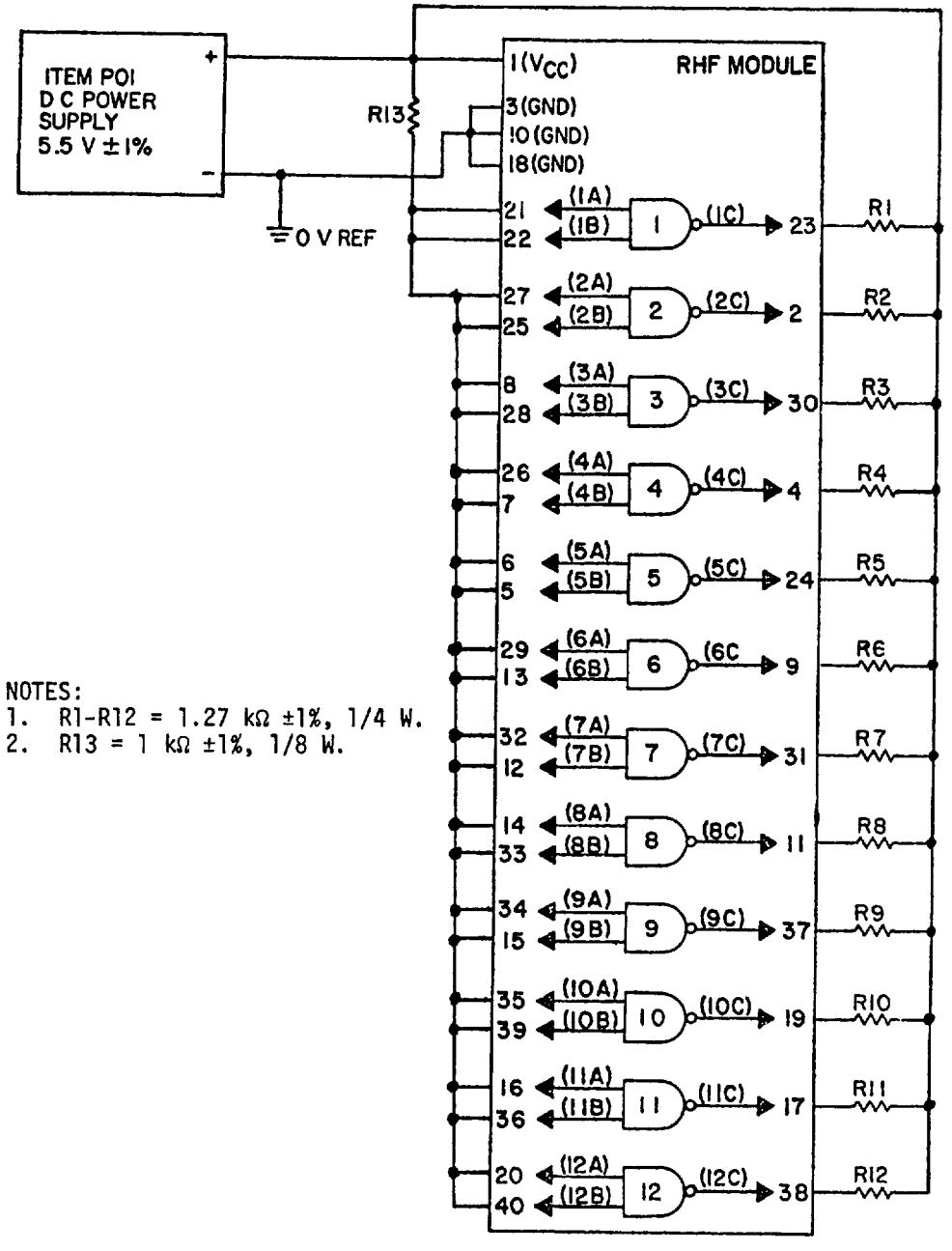
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R10 = 1.27 kΩ ±1%, 1/4 W.
2. R11 = 51Ω ±5%, 1/2 W.
3. R12 = 1 kΩ ±1%, 1/8 W.
4. Item J01 setup:
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.
   PA = 3.5 ±0.5 V.

*FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1-R12 = 1.27 kΩ ±1%, 1/4 W.
2. R13 = 1 kΩ ±1%, 1/8 W.

FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T17:29Z
Check the source to verify that this is the current version before use.

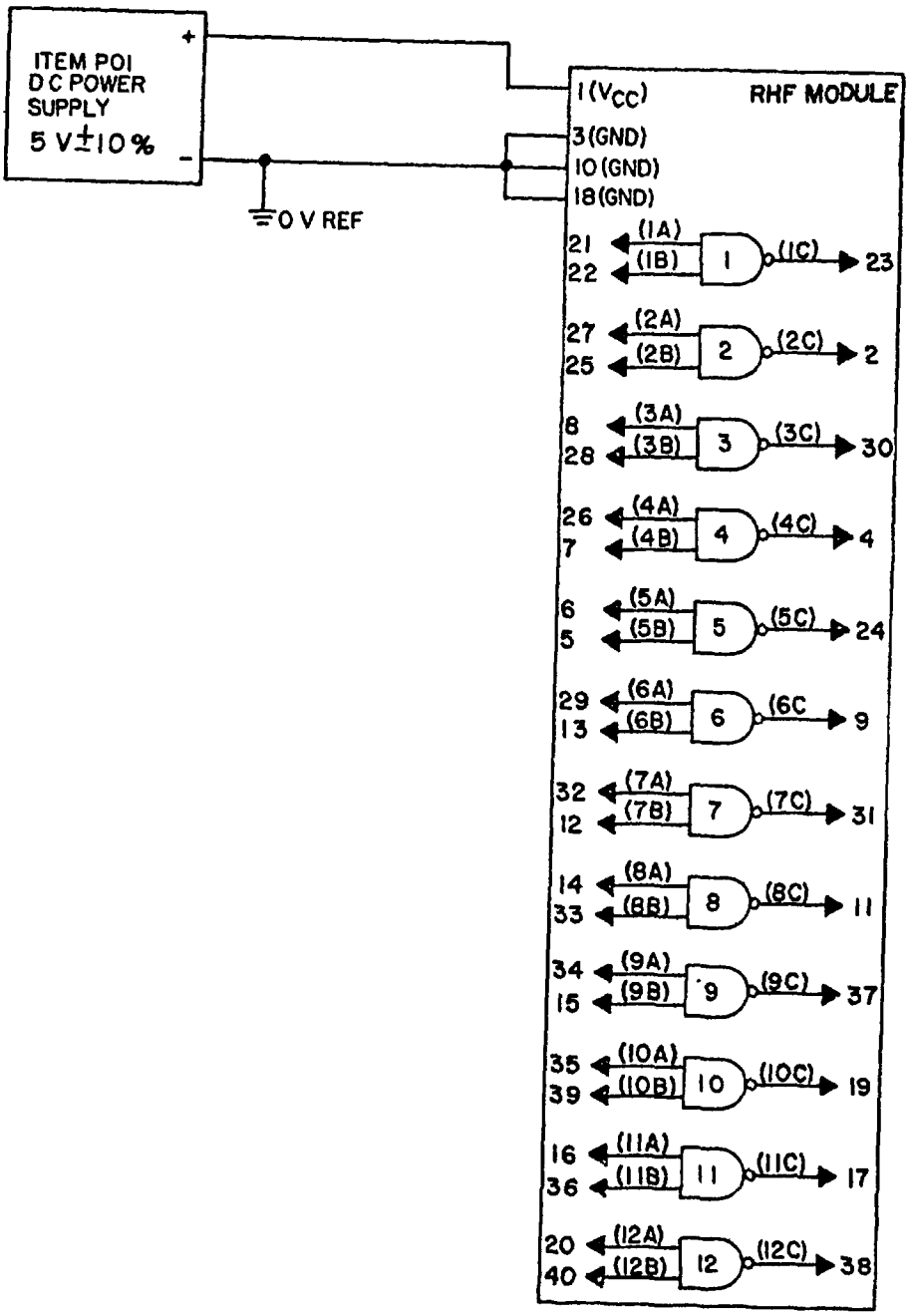


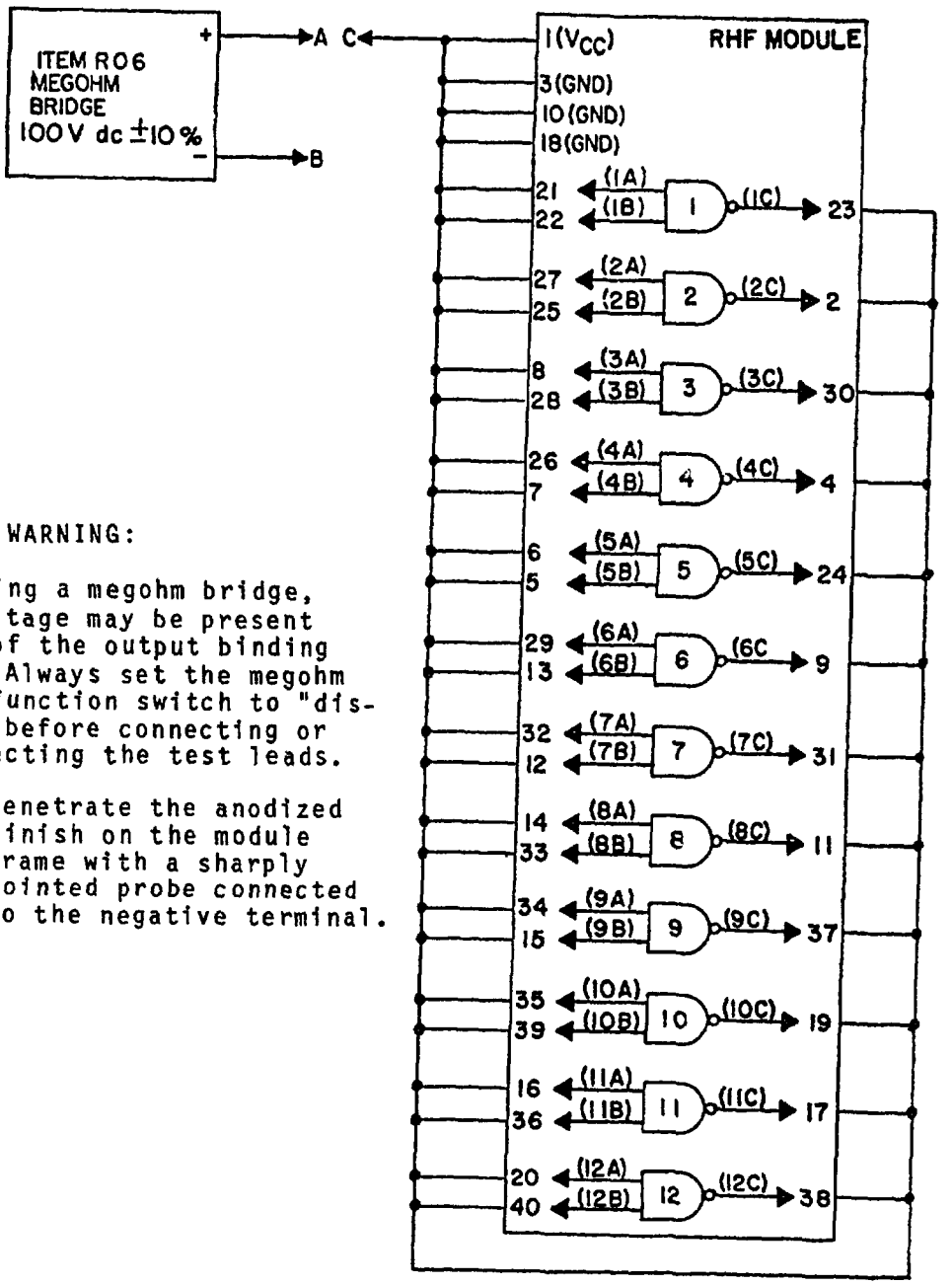
FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	C	Frame	10	∞	MΩ	5

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.



WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

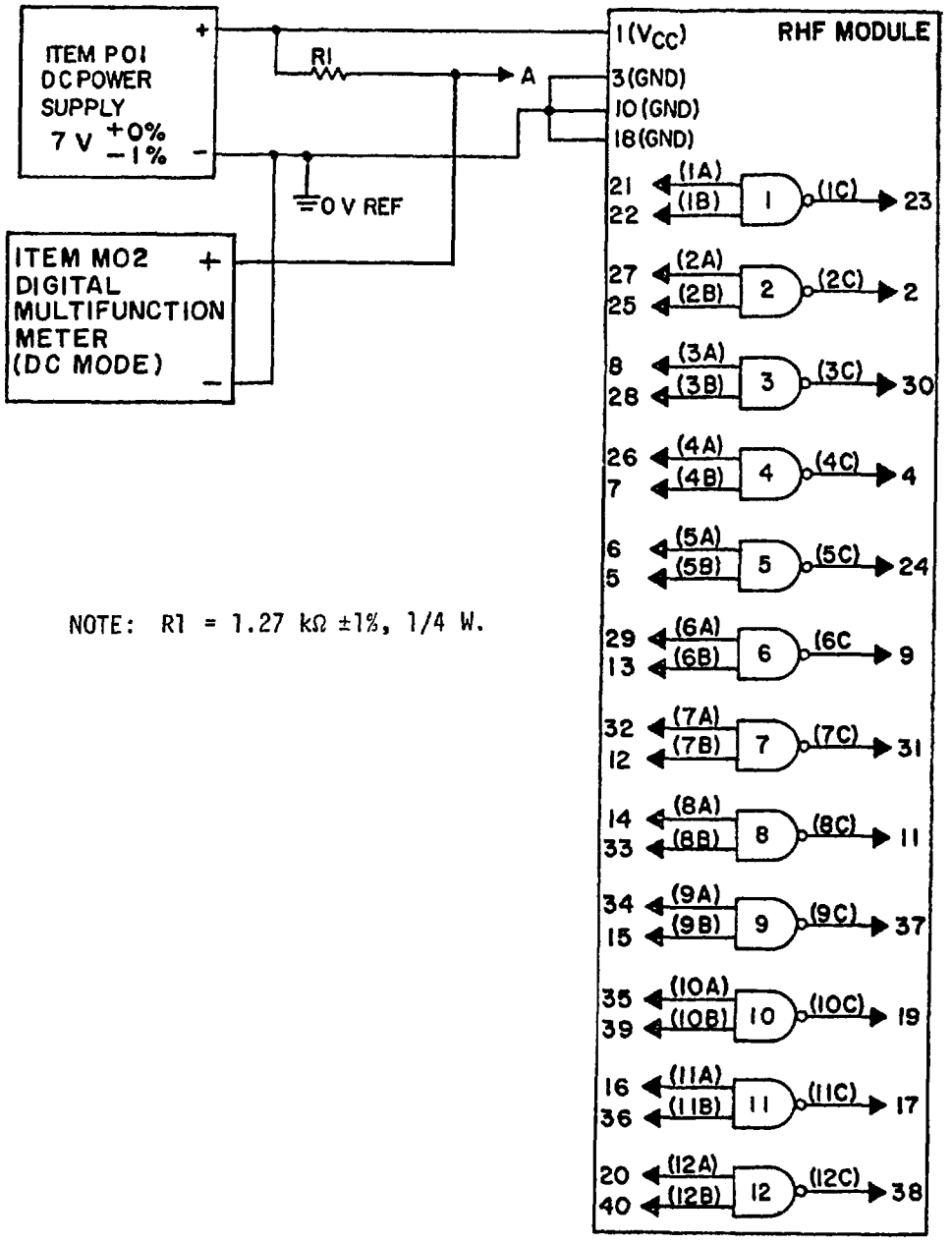
FIGURE 5. Isolation test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 202640-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
B1	1C V_{OUT} (low loaded)	23	10	800	mV	6
B2	2C "	2	"	"	"	"
B3	3C "	30	"	"	"	"
B4	4C "	4	"	"	"	"
B5	5C "	24	"	"	"	"
B6	6C "	9	"	"	"	"
B7	7C "	31	"	"	"	"
B8	8C "	11	"	"	"	"
B9	9C "	37	"	"	"	"
B10	10C "	19	"	"	"	"
B11	11C "	17	"	"	"	"
B12	12C "	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 1.27 kΩ ±1%, 1/4 W.

* FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input breakdown current	5,6,7,8,12,13,14,15,16,20,22, 25,26,27,28,29,32,33,34,35,36, 39,40	21	0	100	µA	7
C2	1B "	5,6,7,8,12,13,14,15,16,20,21, 25,26,27,28,29,32,33,34,35,36, 39,40	22	"	"	"	"
C3	2A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,28,29,32,33,34,35,36, 39,40	27	"	"	"	"
C4	2B "	5,6,7,8,12,13,14,15,16,20,21, 22,26,27,28,29,32,33,34,35,36, 39,40	25	"	"	"	"
C5	3A "	5,6,7,12,13,14,15,16,20,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	8	"	"	"	"
C6	3B "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,29,32,33,34,35,36, 39,40	28	"	"	"	"
C7	4A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,27,28,29,32,33,34,35,36, 39,40	26	"	"	"	"
C8	4B "	5,6,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34,35, 36,39,40	7	"	"	"	"
C9	5A "	5,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34,35, 36,39,40	6	"	"	"	"
C10	5B "	6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34,35, 36,39,40	5	"	"	"	"
C11	6A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,32,33,34,35, 36,39,40	29	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

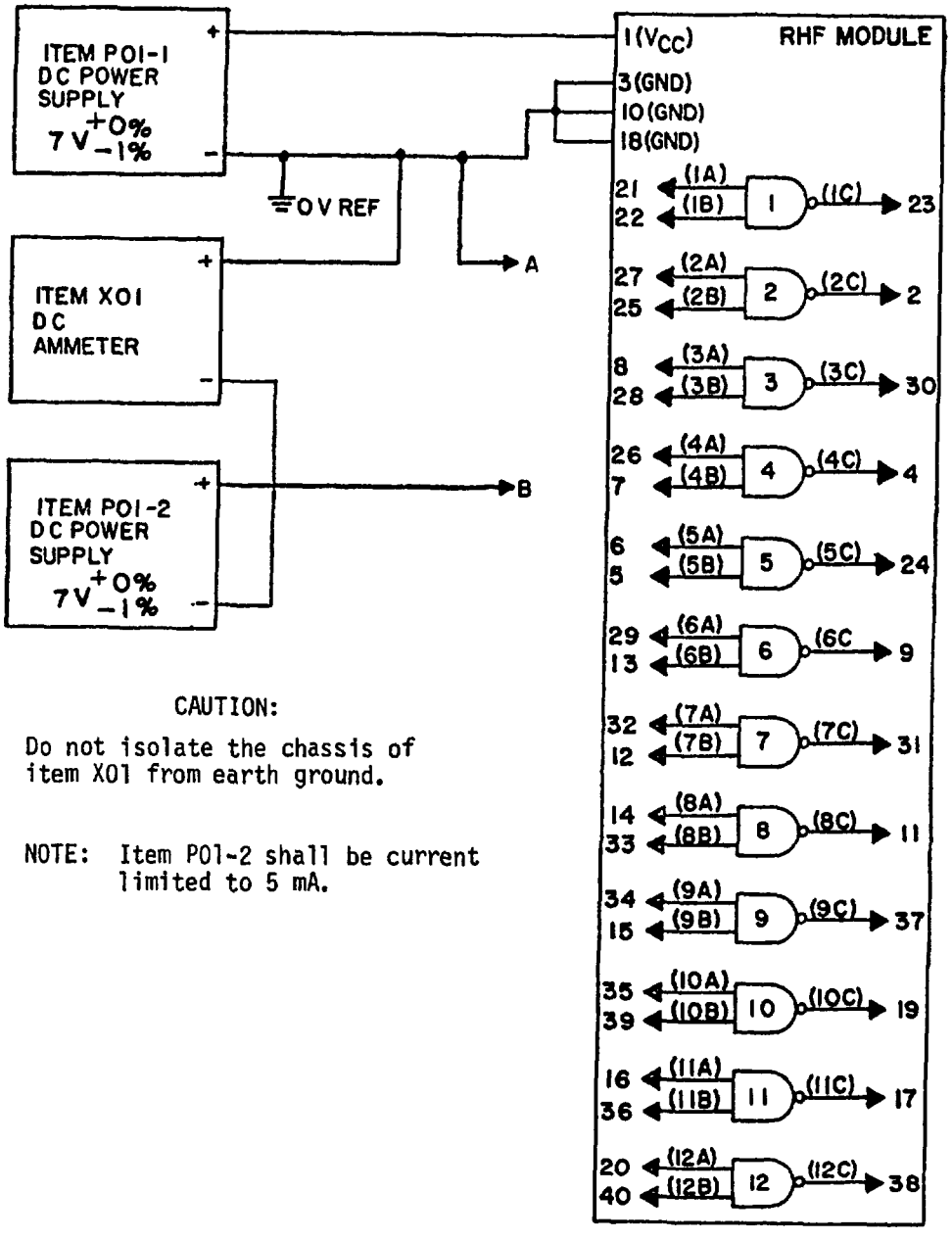
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C12	6B Input breakdown current	5,6,7,8,12,14,15,16,20,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	13	0	100	µA	7
C13	7A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,33,34,35,36, 39,40	32	"	"	"	"
C14	7B "	5,6,7,8,13,14,15,16,20,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	12	"	"	"	"
C15	8A "	5,6,7,8,12,13,15,16,20,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	14	"	"	"	"
C16	8B "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,34,35,36, 39,40	33	"	"	"	"
C17	9A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,35,36, 39,40	34	"	"	"	"
C18	9B "	5,6,7,8,12,13,14,16,20,21,22, 25,26,27,28,29,32,33,34,35, 36,39,40	15	"	"	"	"
C19	10A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 36,39,40	35	"	"	"	"
C20	10B "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,40	39	"	"	"	"
C21	11A "	5,6,7,8,12,13,14,15,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	16	"	"	"	"
C22	11B "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,39,40	36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C23	12A Input breakdown current	5,6,7,8,12,13,14,15,16,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	20	0	100	µA	7
C24	12B Input breakdown current	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34,35, 36,39	40	0	100	µA	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: Item P01-2 shall be current limited to 5 mA.

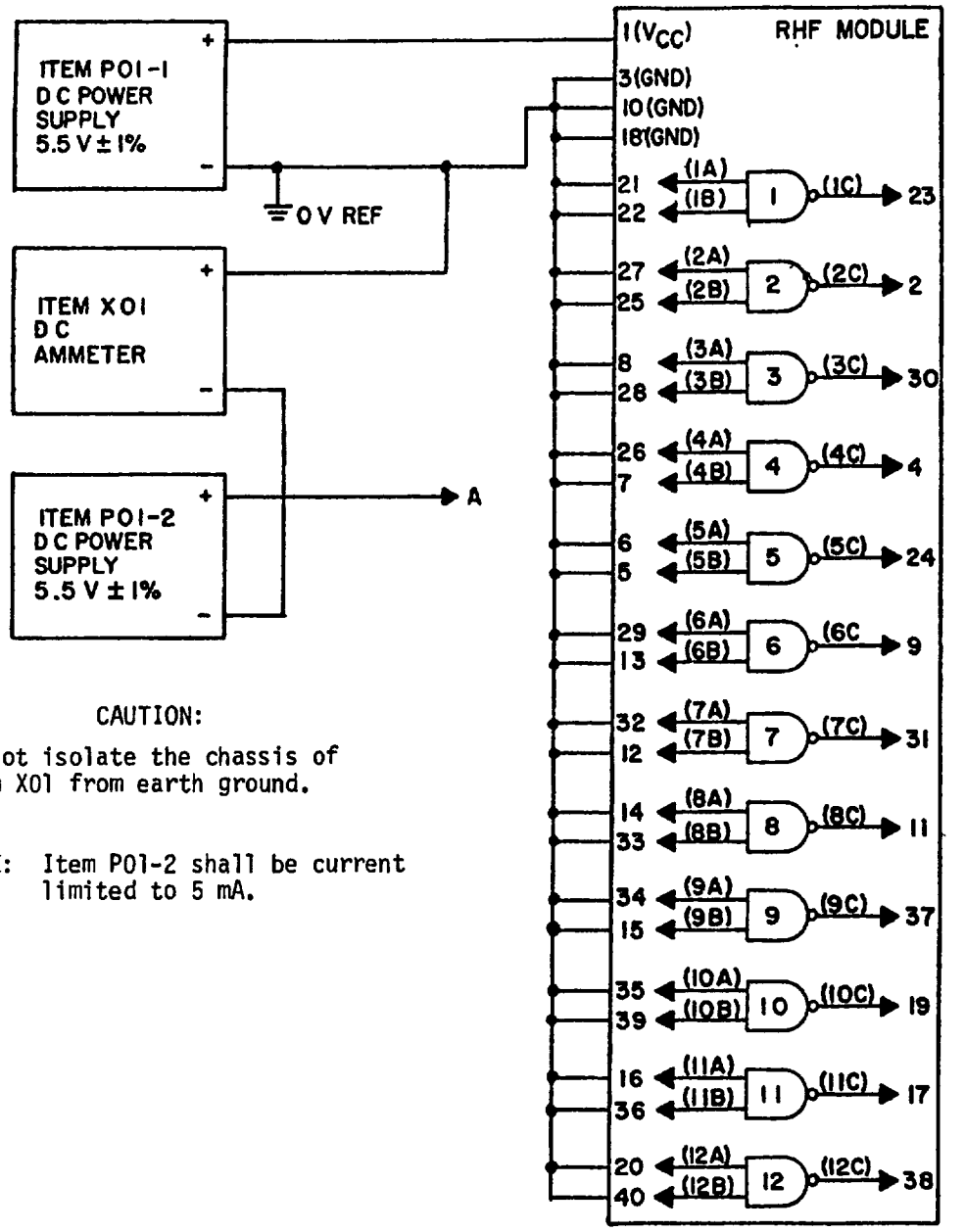
FIGURE 7. Input breakdown current test circuit for table V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE VI. Output leakage current.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
D1	1C Output leakage current	23	0	1	mA	8
D2	2C "	2	"	"	"	"
D3	3C "	30	"	"	"	"
D4	4C "	4	"	"	"	"
D5	5C "	24	"	"	"	"
D6	6C "	9	"	"	"	"
D7	7C "	31	"	"	"	"
D8	8C "	11	"	"	"	"
D9	9C "	37	"	"	"	"
D10	10C "	19	"	"	"	"
D11	11C "	17	"	"	"	"
D12	12C "	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: Item P01-2 shall be current limited to 5 mA.

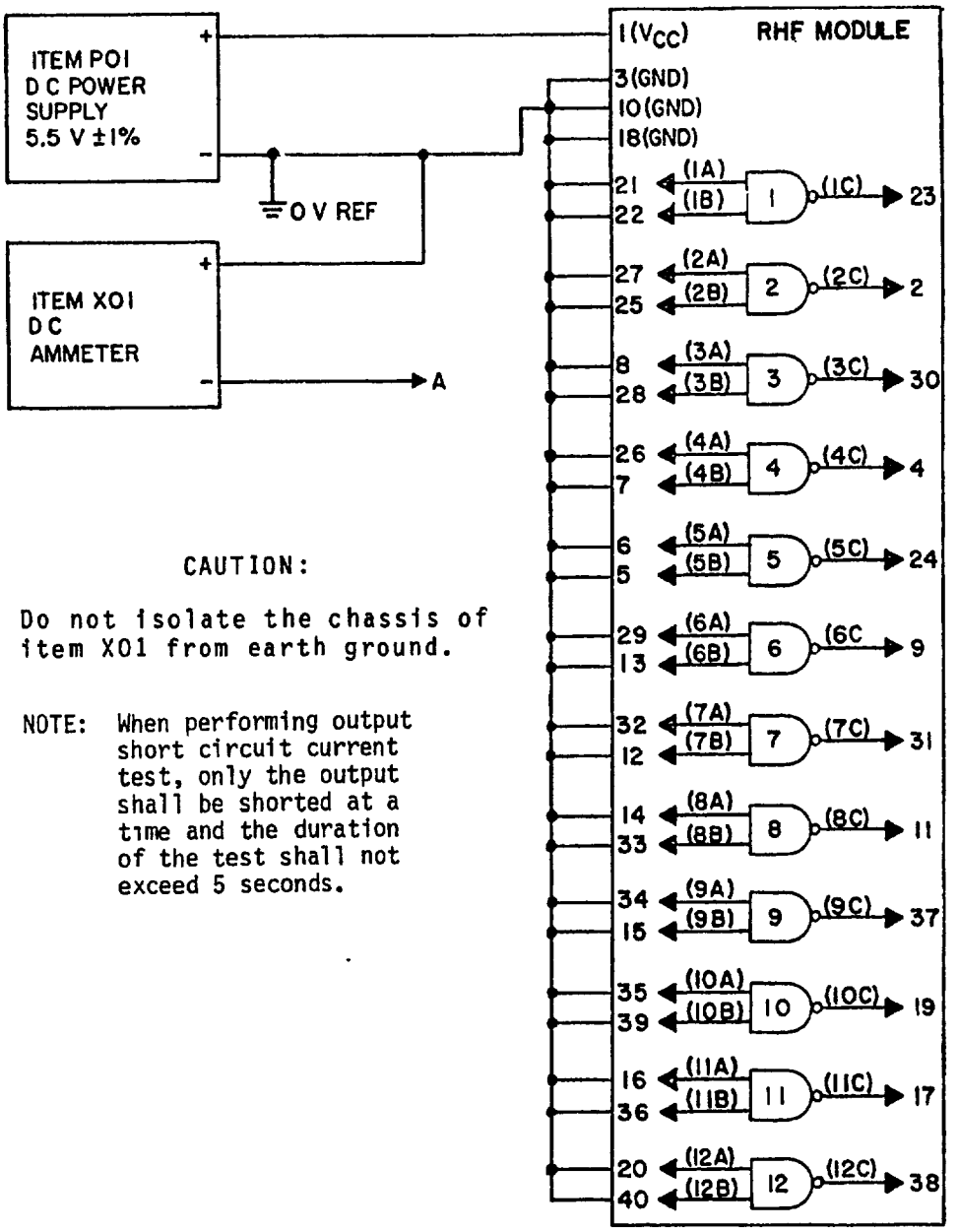
FIGURE 8. Output leakage current test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE VII. Output short circuit current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1C Output short circuit current	23	-15	-100	mA	9
E2	2C "	2	"	"	"	"
E3	3C "	30	"	"	"	"
E4	4C "	4	"	"	"	"
E5	5C "	24	"	"	"	"
E6	6C "	9	"	"	"	"
E7	7C "	31	"	"	"	"
E8	8C "	11	"	"	"	"
E9	9C "	37	"	"	"	"
E10	10C "	19	"	"	"	"
E11	11C "	17	"	"	"	"
E12	12C "	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: When performing output short circuit current test, only the output shall be shorted at a time and the duration of the test shall not exceed 5 seconds.

FIGURE 9. Output short circuit current test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input clamp diode voltage test.

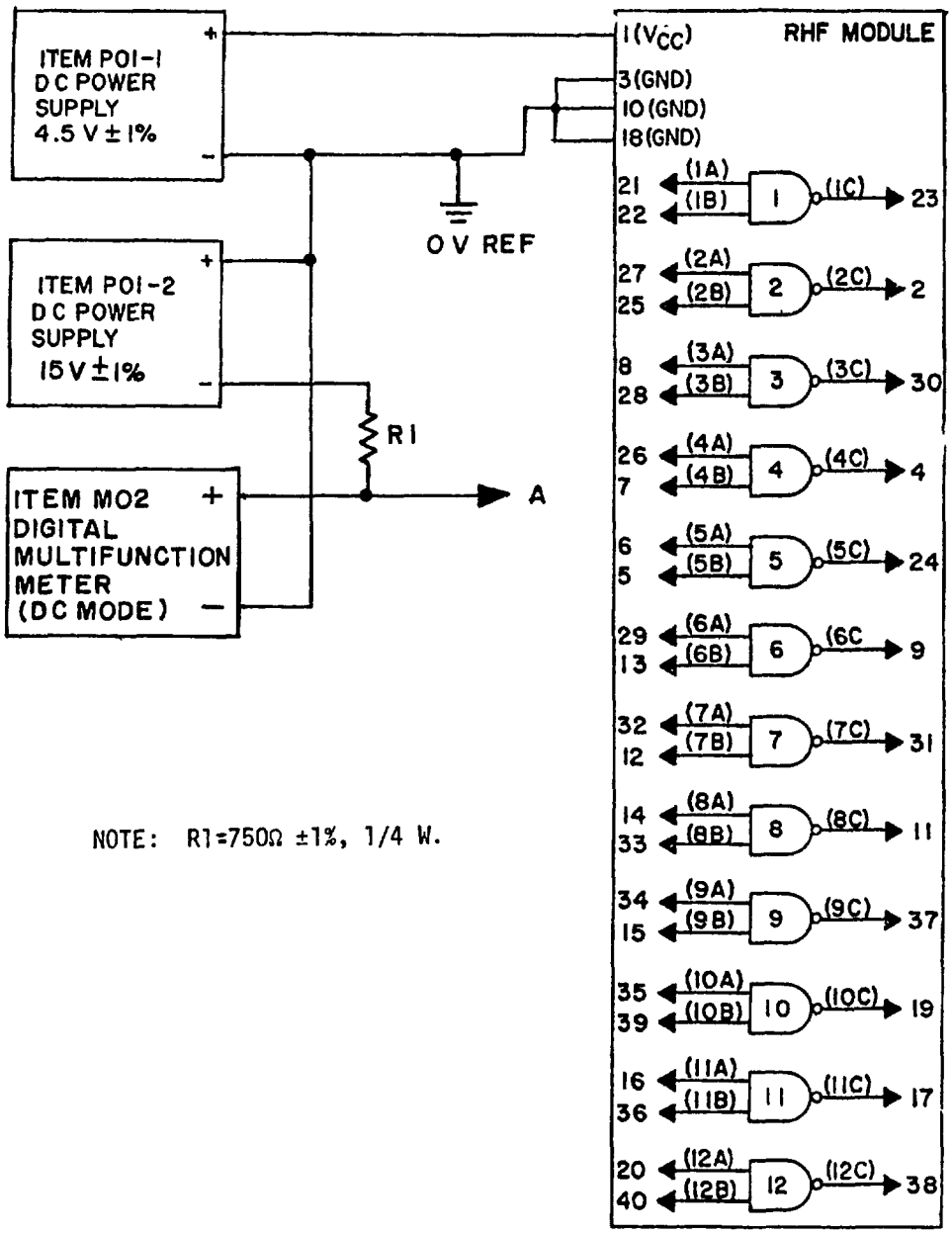
Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
F1	1A Input clamp diode voltage	21	-0.4	-1.5	V	10
F2	1B "	22	"	"	"	"
F3	2A "	27	"	"	"	"
F4	2B "	25	"	"	"	"
F5	3A "	8	"	"	"	"
F6	3B "	28	"	"	"	"
F7	4A "	26	"	"	"	"
F8	4B "	7	"	"	"	"
F9	5A "	6	"	"	"	"
F10	5B "	5	"	"	"	"
F11	6A "	29	"	"	"	"
F12	6B "	13	"	"	"	"
F13	7A "	32	"	"	"	"
F14	7B "	12	"	"	"	"
F15	8A "	14	"	"	"	"
F16	8B "	33	"	"	"	"
F17	9A "	34	"	"	"	"
F18	9B "	15	"	"	"	"
F19	10A "	35	"	"	"	"
F20	10B "	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F21	11A Input clamp diode voltage	16	-0.4	-1.5	V	10
F22	11B "	36	"	"	"	"
F23	12A "	20	"	"	"	"
F24	12B "	40	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTE: R1=750Ω ±1%, 1/4 W.

*FIGURE 10. Input clamp diode voltage test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE IX. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1C Output voltage, low level	21,22	Open	23	0	400	mV	11
G2	2C "	25,27	"	2	"	"	"	"
G3	3C "	8,28	"	30	"	"	"	"
G4	4C "	7,26	"	4	"	"	"	"
G5	5C "	5,6	"	24	"	"	"	"
G6	6C "	13,29	"	9	"	"	"	"
G7	7C "	12,32	"	31	"	"	"	"
G8	8C "	14,33	"	11	"	"	"	"
G9	9C "	15,34	"	37	"	"	"	"
G10	10C "	35,39	"	19	"	"	"	"
G11	11C "	16,36	"	17	"	"	"	"
G12	12C "	20,40	"	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE X. Output voltage (high-level) test.

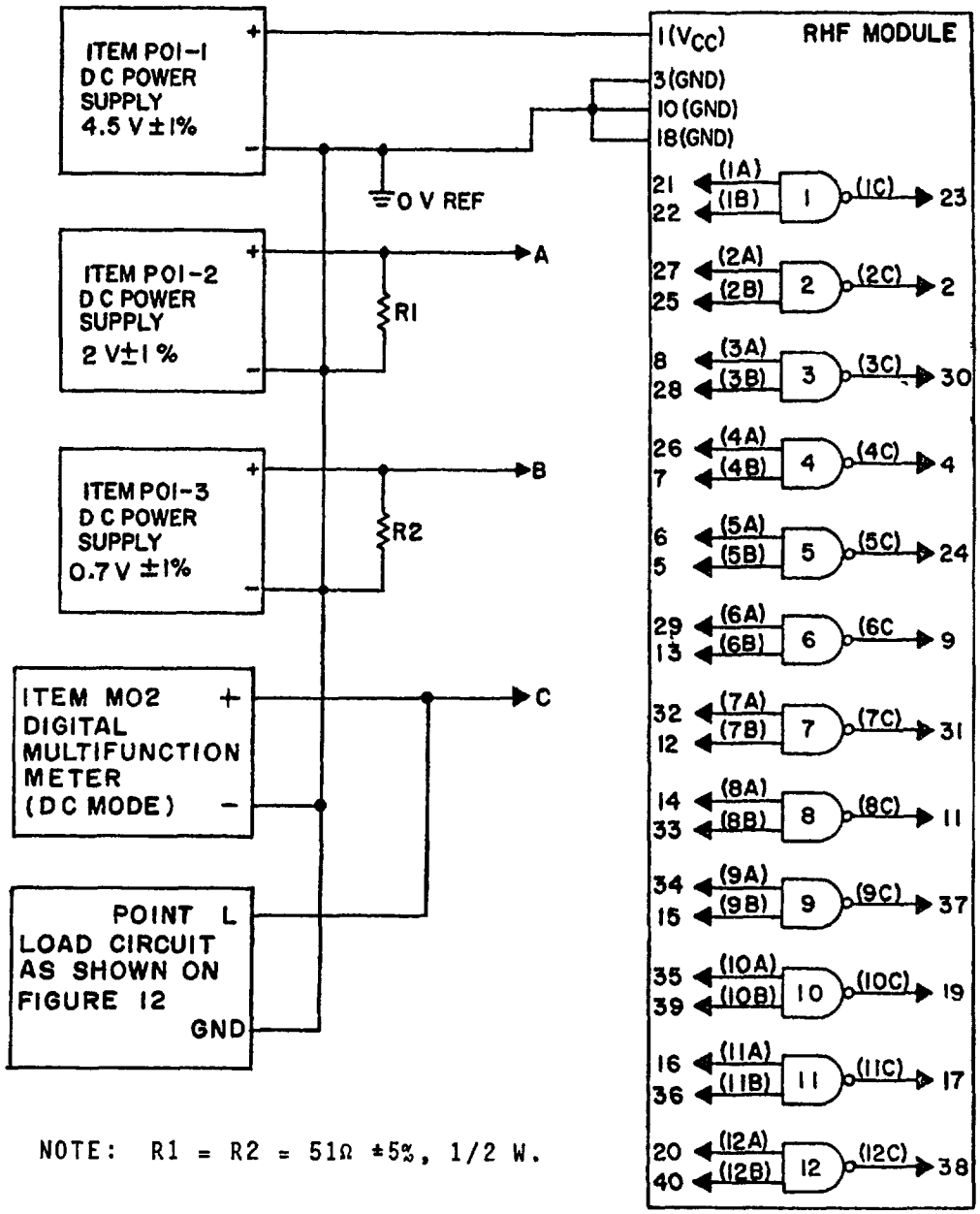
Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H1	1C Output voltage, high level	22	21	23	2.5	4.5	V	11
H2	2C "	25	27	2	"	"	"	"
H3	3C "	28	8	30	"	"	"	"
H4	4C "	7	26	4	"	"	"	"
H5	5C "	5	6	24	"	"	"	"
H6	6C "	13	29	9	"	"	"	"
H7	7C "	12	32	31	"	"	"	"
H8	8C "	33	14	11	"	"	"	"
H9	9C "	15	34	37	"	"	"	"
H10	10C "	39	35	19	"	"	"	"
H11	11C "	36	16	17	"	"	"	"
H12	12C "	40	20	38	"	"	"	"
H13	1C "	21	22	23	"	"	"	"
H14	2C "	27	25	2	"	"	"	"
H15	3C "	8	28	30	"	"	"	"
H16	4C "	26	7	4	"	"	"	"
H17	5C "	6	5	24	"	"	"	"
H18	6C "	29	13	9	"	"	"	"
H19	7C "	32	12	31	"	"	"	"
H20	8C "	14	33	11	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE X. Output voltage (high-level) test - Continued.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
H21	9C Output voltage, high level	34	15	37	2.5	4.5	V	11
H22	10C "	35	39	19	"	"	"	"
H23	11C "	16	36	17	"	"	"	"
H24	12C "	20	40	38	"	"	"	"

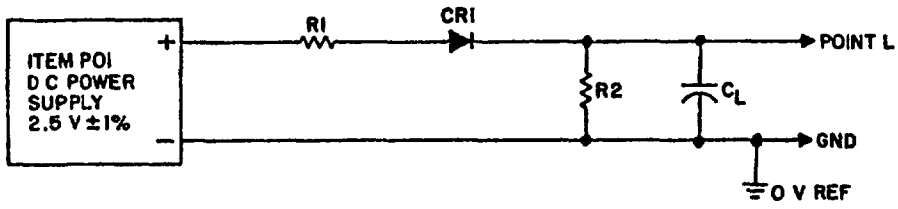
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

*FIGURE 11. Output voltage (low-level and high-level) test circuit for tables IX and X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 4 mA ±5% out of point L with point L held at 0.4 V.
3. R2 = 6.19 kΩ ±1%, 1/8 W.
4. C_L = 50 ±5 pF (including probe and parasitic capacitance) for test conditions N1-N12, P1-P12, Q1-Q12, R1-R12, S1-S12, and T1-T12.
5. C_L = 150 ±5 pF (including probe and parasitic capacitance) for test conditions N13-N24, P13-P24, Q13-Q24, and R13-R24.
6. Capacitor C_L is not required for dc electrical test conditions G1-G12 and H1-H24.
7. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

FIGURE 12. RHF logic load circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A Input current, high level	5,6,7,8,12,13,14,15,16,20,22, 25,26,27,28,29,32,33,34,35,36, 39,40	21	0	20	µA	13
J2	1B "	5,6,7,8,12,13,14,15,16,20,21, 25,26,27,28,29,32,33,34,35,36, 39,40	22	"	"	"	"
J3	2A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,28,29,32,33,34,35,36, 39,40	27	"	"	"	"
J4	2B "	5,6,7,8,12,13,14,15,16,20,21, 22,26,27,28,29,32,33,34,35,36, 39,40	25	"	"	"	"
J5	3A "	5,6,7,12,13,14,15,16,20,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	8	"	"	"	"
J6	3B "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,29,32,33,34,35,36, 39,40	28	"	"	"	"
J7	4A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,27,28,29,32,33,34,35,36, 39,40	26	"	"	"	"
J8	4B "	5,6,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34,35, 36,39,40	7	"	"	"	"
J9	5A "	5,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34,35, 36,39,40	6	"	"	"	"
J10	5B "	6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34,35, 36,39,40	5	"	"	"	"
J11	6A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,32,33,34,35, 36,39,40	29	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test - Continued.

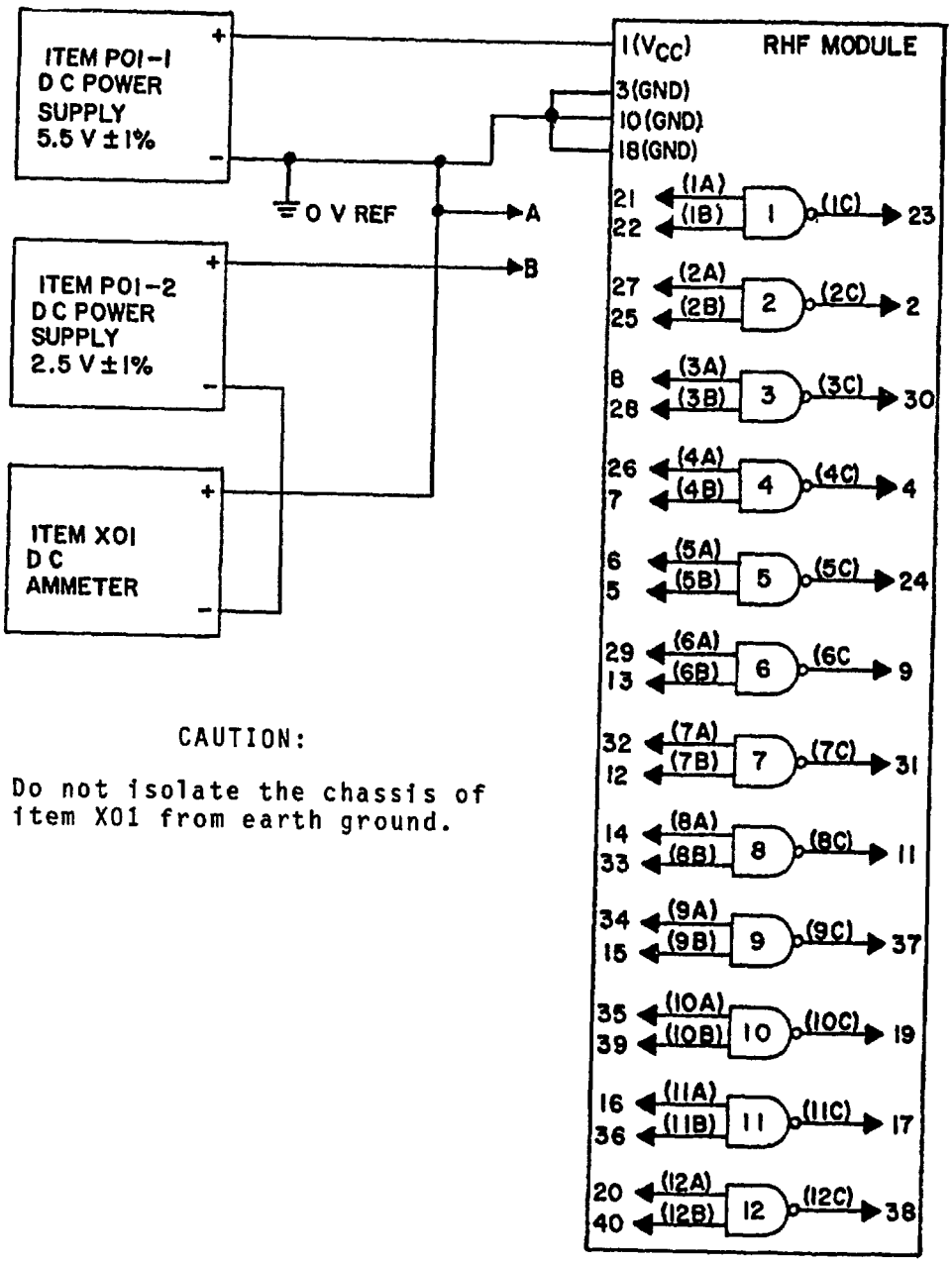
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J12	6B Input current, high level	5,6,7,8,12,14,15,16,20,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	13	0	20	µA	13
J13	7A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,33,34,35,36, 39,40	32	"	"	"	"
J14	7B "	5,6,7,8,13,14,15,16,20,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	12	"	"	"	"
J15	8A "	5,6,7,8,12,13,15,16,20,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	14	"	"	"	"
J16	8B "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,34,35,36, 39,40	33	"	"	"	"
J17	9A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,35,36, 39,40	34	"	"	"	"
J18	9B "	5,6,7,8,12,13,14,16,20,21,22, 25,26,27,28,29,32,33,34,35, 36,39,40	15	"	"	"	"
J19	10A "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 36,39,40	35	"	"	"	"
J20	10B "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,36,40	39	"	"	"	"
J21	11A "	5,6,7,8,12,13,14,15,20,21, 22,25,26,27,28,29,32,33,34, 35,36,39,40	16	"	"	"	"
J22	11B "	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34, 35,39,40	36	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J23	12A Input current, high level	5,6,7,8,12,13,14,15,16,21,22, 25,26,27,28,29,32,33,34,35,36, 39,40	20	0	20	µA	13
J24	12B Input current, high level	5,6,7,8,12,13,14,15,16,20,21, 22,25,26,27,28,29,32,33,34,35, 36,39	40	0	20	µA	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 13. Input current (high-level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE XII. Input current (low-level) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
K1	1A Input current, low level	21	-10	-430	µA	14
K2	1B "	22	"	"	"	"
K3	2A "	27	"	"	"	"
K4	2B "	25	"	"	"	"
K5	3A "	8	"	"	"	"
K6	3B "	28	"	"	"	"
K7	4A "	26	"	"	"	"
K8	4B "	7	"	"	"	"
K9	5A "	6	"	"	"	"
K10	5B "	5	"	"	"	"
K11	6A "	29	"	"	"	"
K12	6B "	13	"	"	"	"
K13	7A "	32	"	"	"	"
K14	7B "	12	"	"	"	"
K15	8A "	14	"	"	"	"
K16	8B "	33	"	"	"	"
K17	9A "	34	"	"	"	"
K18	9B "	15	"	"	"	"
K19	10A "	35	"	"	"	"
K20	10B "	39	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
K21	11A Input current, low level	16	-10	-430	µA	14
K22	11B "	36	"	"	"	"
K23	12A "	20	"	"	"	"
K24	12B "	40	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

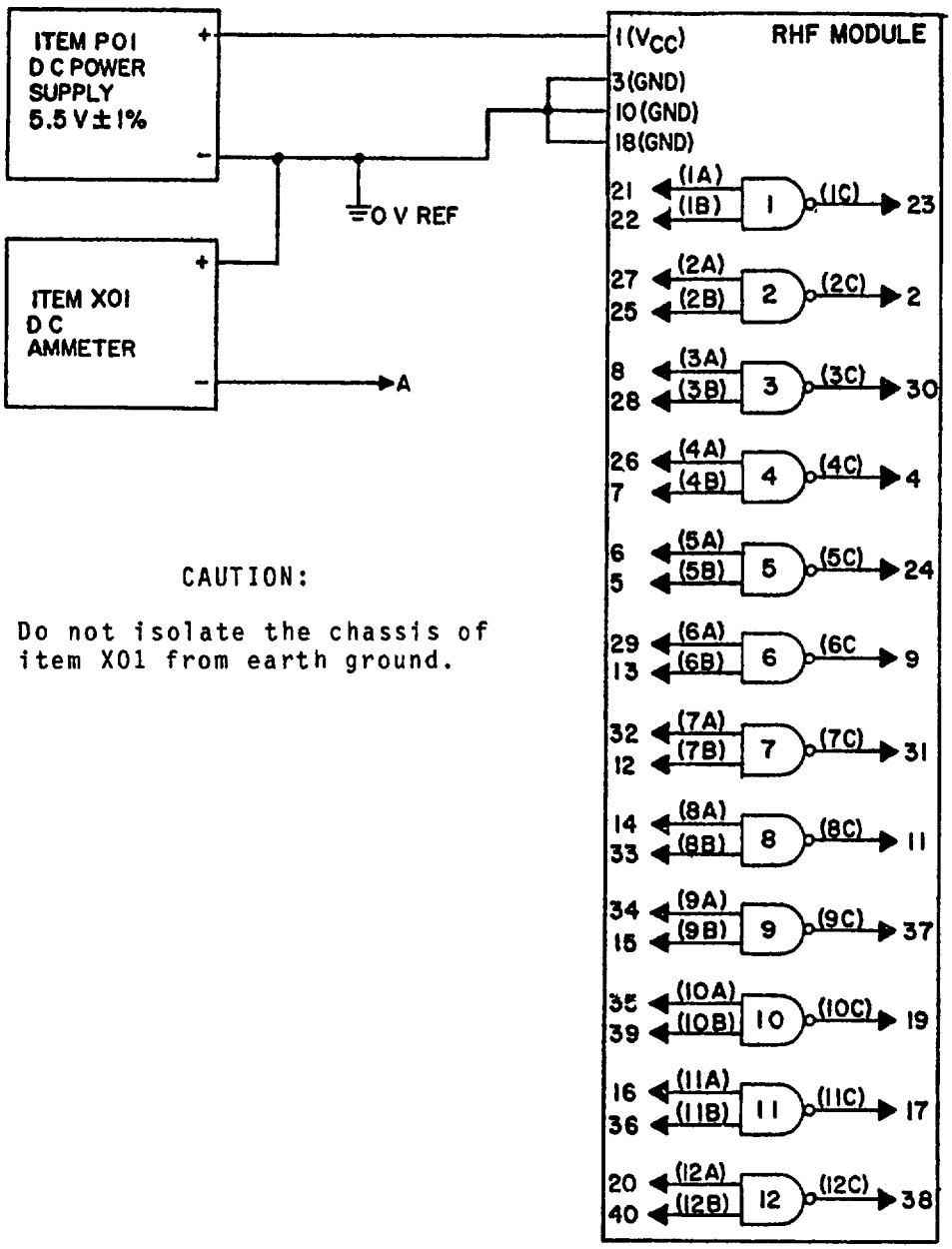


FIGURE 14. Input current (low-level) test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XIII. Ground pin continuity test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Ground pin continuity	3	0	1	Ω	15
L2	"	18	0	1	"	"
L3	"	3	0.00	0.10	"	"
L4	"	18	0.00	0.10	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

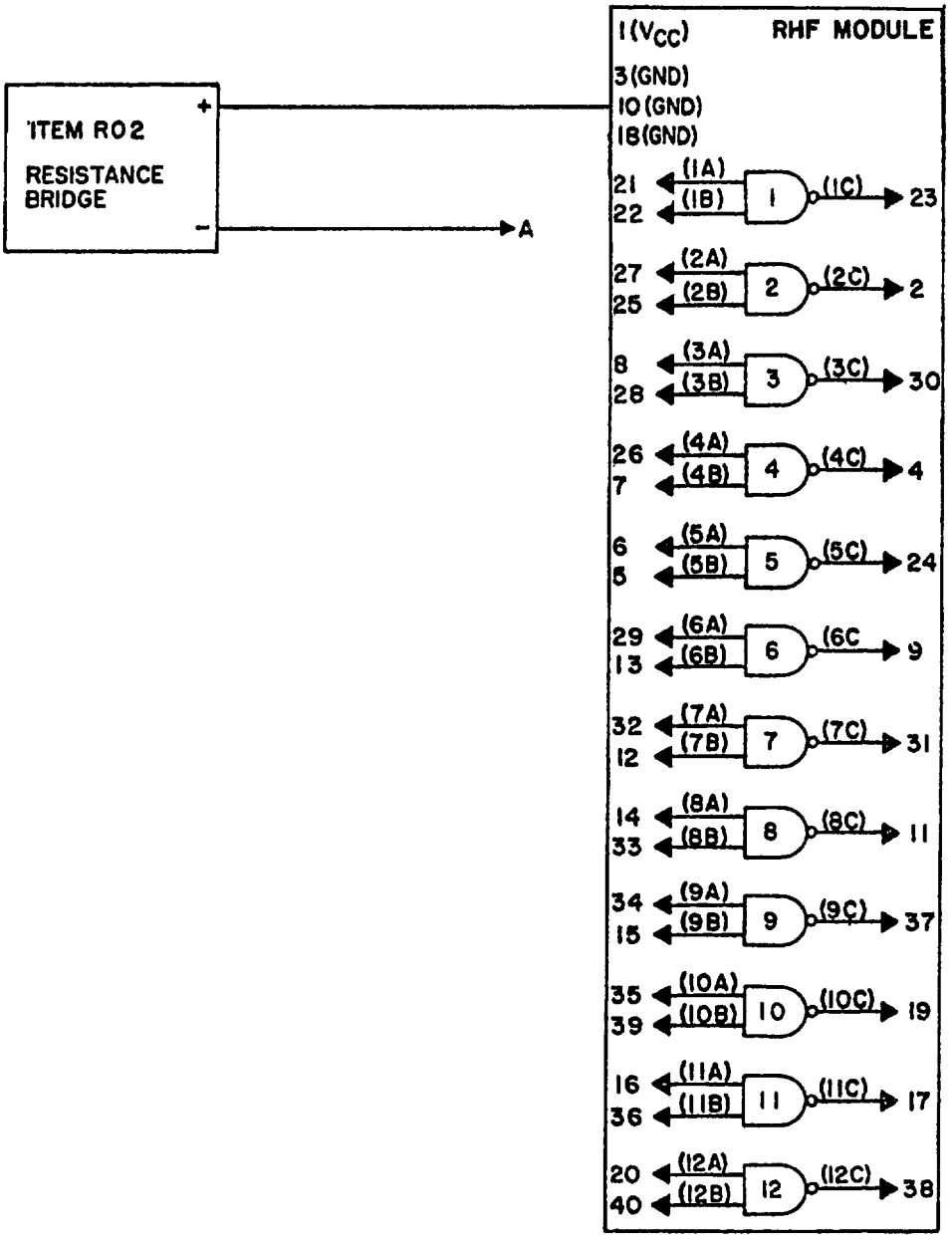


FIGURE 15. Ground pin continuity test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XIV. Power supply current (low-level outputs and high-level outputs) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
M1	Power supply current, low level outputs	C	Open	0.1	17.6	mA	16
M2	Power supply current, high level outputs	Open	C	0.1	6.4	mA	16

Source: https://assist.dla.mil -- Downloaded: 20261O-20T17:29Z
Check the source to verify that this is the current version before use.

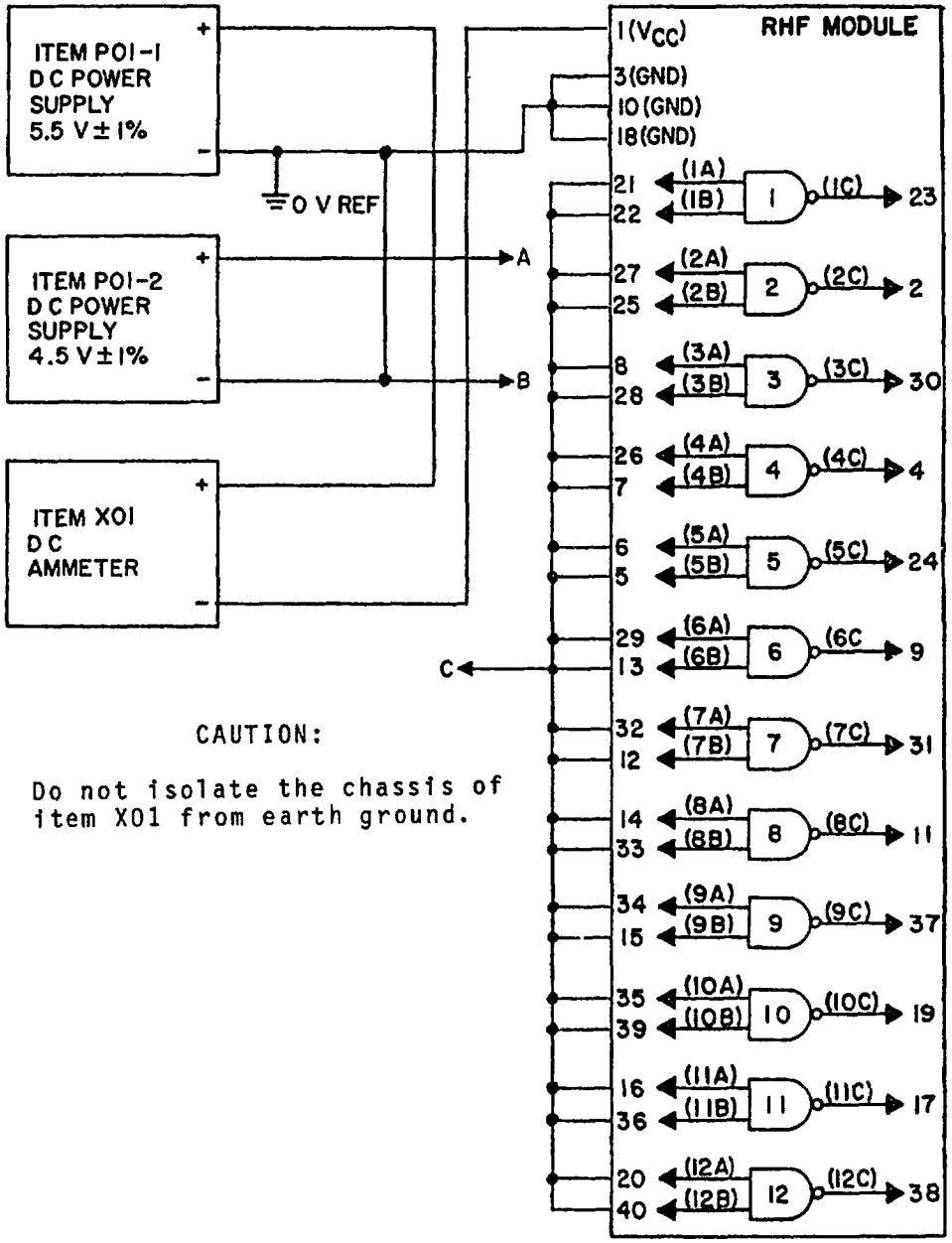


FIGURE 16. Power supply current (low-level outputs and high-level outputs) test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N1	1C Transition time (t_{TLH}), C_L = 50 pF	21	22	23	Open	5	50	ns	17,18
N2	2C "	27	25	2	"	"	"	"	"
N3	3C "	8	28	30	"	"	"	"	"
N4	4C "	26	7	4	"	"	"	"	"
N5	5C "	6	5	24	"	"	"	"	"
N6	6C "	29	13	9	"	"	"	"	"
N7	7C "	32	12	31	"	"	"	"	"
N8	8C "	14	33	11	"	"	"	"	"
N9	9C "	34	15	37	"	"	"	"	"
N10	10C "	35	39	19	"	"	"	"	"
N11	11C "	16	36	17	"	"	"	"	"
N12	12C "	20	40	38	"	"	"	"	"
N13	1C Transition time (t_{TLH}), C_L = 150 pF	21	22	23	"	10	80	"	"
N14	2C "	27	25	2	"	"	"	"	"
N15	3C "	8	28	30	"	"	"	"	"
N16	4C "	26	7	4	"	"	"	"	"
N17	5C "	6	5	24	"	"	"	"	"
N18	6C "	29	13	9	"	"	"	"	"
N19	7C "	32	12	31	"	"	"	"	"
N20	8C "	14	33	11	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
N21	9C Transition time (t_{TLH}), C_L = 150 pF	34	15	37	Open	10	80	ns	17,18
N22	10C "	35	39	19	"	"	"	"	"
N23	11C "	16	36	17	"	"	"	"	"
N24	12C "	20	40	38	"	"	"	"	"
P1	1C Transition time (t_{THL}), C_L = 50 pF	22	21	23	5,6,7,8,12,13,14,15, 16,20,25,26,27,28,29, 32,33,34,35,36,39,40	3	18	"	"
P2	2C "	25	27	2	5,6,7,8,12,13,14,15, 16,20,21,22,26,28,29, 32,33,34,35,36,39,40	"	"	"	"
P3	3C "	28	8	30	5,6,7,12,13,14,15,16, 20,21,22,25,26,27,29, 32,33,34,35,36,39,40	"	"	"	"
P4	4C "	7	26	4	5,6,8,12,13,14,15,16, 20,21,22,25,27,28,29, 32,33,34,35,36,39,40	"	"	"	"
P5	5C "	5	6	24	7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,39, 40	"	"	"	"
P6	6C "	13	29	9	5,6,7,8,12,14,15,16, 20,21,22,25,26,27,28, 32,33,34,35,36,39,40	"	"	"	"
P7	7C "	12	32	31	5,6,7,8,13,14,15,16, 20,21,22,25,26,27,28, 29,33,34,35,36,39,40	"	"	"	"
P8	8C "	33	14	11	5,6,7,8,12,13,15,16, 20,21,22,25,26,27,28, 29,32,34,35,36,39,40	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P9	9C Transition time (t_{THL}), C_L = 50 pF	15	34	37	5,6,7,8,12,13,14,16, 20,21,22,25,26,27,28, 29,32,33,35,36,39,40	3	18	ns	17,18
P10	10C "	39	35	19	5,6,7,8,12,13,14,15, 16,20,21,22,25,26,27, 28,29,32,33,34,36,40	"	"	"	"
P11	11C "	36	16	17	5,6,7,8,12,13,14,15, 20,21,22,25,26,27,28, 29,32,33,34,35,39,40	"	"	"	"
P12	12C "	40	20	38	5,6,7,8,12,13,14,15, 16,21,22,25,26,27,28, 29,32,33,34,35,36,39	"	"	"	"
P13	1C Transition time (t_{THL}), C_L = 150 pF	22	21	23	5,6,7,8,12,13,14,15, 16,20,25,26,27,28,29, 32,33,34,35,36,39,40	6	40	"	"
P14	2C "	25	27	2	5,6,7,8,12,13,14,15, 16,20,21,22,26,28,29, 32,33,34,35,36,39,40	"	"	"	"
P15	3C "	28	8	30	5,6,7,12,13,14,15,16, 20,21,22,25,26,27,29, 32,33,34,35,36,39,40	"	"	"	"
P16	4C "	7	26	4	5,6,8,12,13,14,15,16, 20,21,22,25,27,28,29, 32,33,34,35,36,39,40	"	"	"	"
P17	5C "	5	6	24	7,8,12,13,14,15,16, 20,21,22,25,26,27,28, 29,32,33,34,35,36,39, 40	"	"	"	"
P18	6C "	13	29	9	5,6,7,8,12,14,15,16, 20,21,22,25,26,27,28, 32,33,34,35,36,39,40	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
P19	7C Transition time (t_{THL}), C_L = 150 pF	12	32	31	5,6,7,8,13,14,15,16, 20,21,22,25,26,27,28, 29,33,34,35,36,39,40	6	40	ns	17,18
P20	8C "	33	14	11	5,6,7,8,12,13,15,16, 20,21,22,25,26,27,28, 29,32,34,35,36,39,40	"	"	"	"
P21	9C "	15	34	37	5,6,7,8,12,13,14,16, 20,21,22,25,26,27,28, 29,32,33,35,36,39,40	"	"	"	"
P22	10C "	39	35	19	5,6,7,8,12,13,14,15, 16,20,21,22,25,26,27, 28,29,32,33,34,36,40	"	"	"	"
P23	11C "	36	16	17	5,6,7,8,12,13,14,15, 20,21,22,25,26,27,28, 29,32,33,34,35,39,40	"	"	"	"
P24	12C "	40	20	38	5,6,7,8,12,13,14,15, 16,21,22,25,26,27,28, 29,32,33,34,35,36,39	"	"	"	"
Q1	1C Propagation delay time (t_{PLH}), C_L = 50 pF	21	22	23	Open	3	19	"	"
Q2	2C "	27	25	2	"	"	"	"	"
Q3	3C "	8	28	30	"	"	"	"	"
Q4	4C "	26	7	4	"	"	"	"	"
Q5	5C "	6	5	24	"	"	"	"	"
Q6	6C "	29	13	9	"	"	"	"	"
Q7	7C "	32	12	31	"	"	"	"	"
Q8	8C "	14	33	11	"	"	"	"	"
Q9	9C "	34	15	37	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

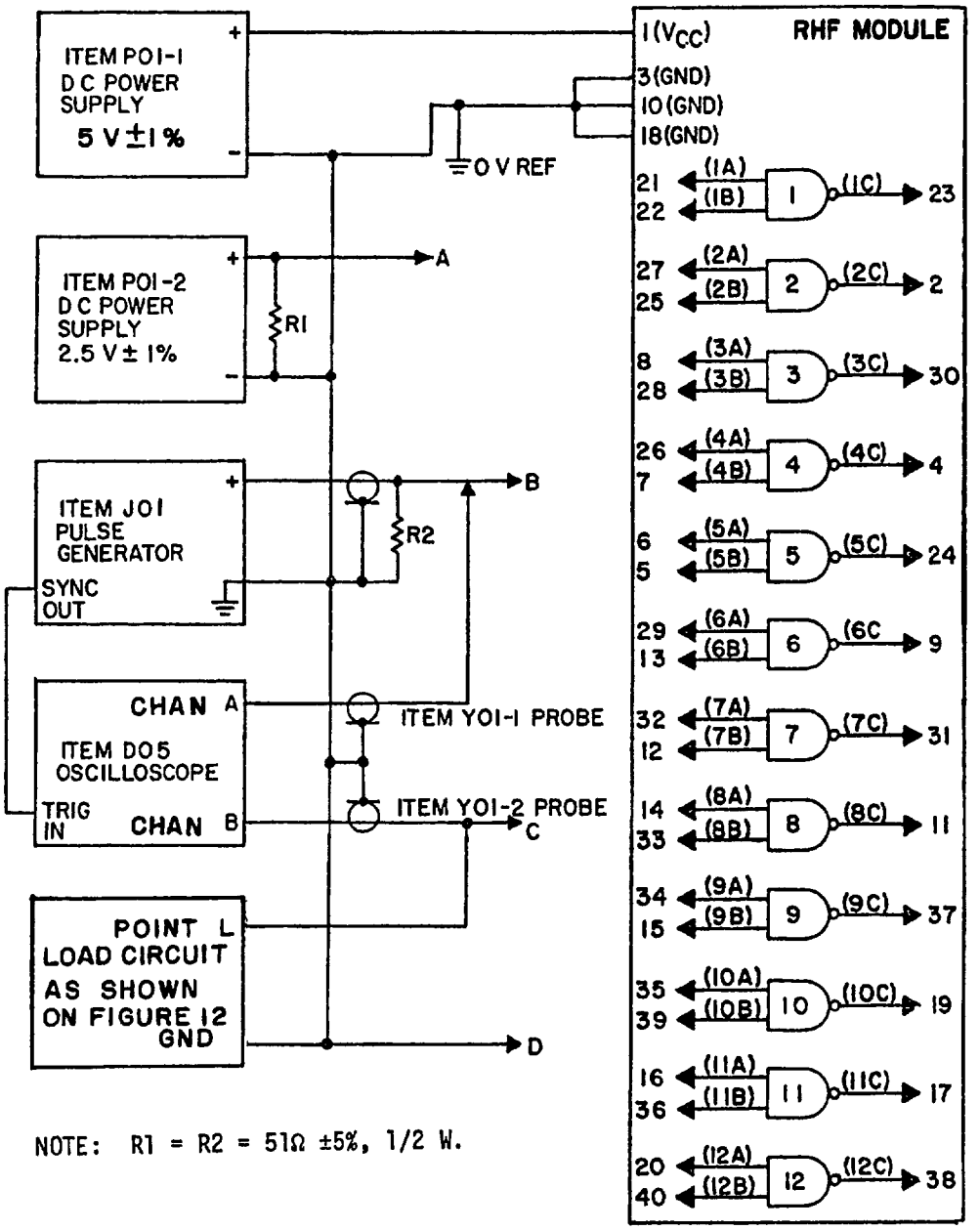
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
Q10	10C Propagation delay time (t_{PLH}), C_L = 50 pF	35	39	19	Open	3	19	ns	17,18
Q11	11C "	16	36	17	"	"	"	"	"
Q12	12C "	20	40	38	"	"	"	"	"
Q13	1C Propagation delay time (t_{PLH}), C_L = 150 pF	21	22	23	"	4	29	"	"
Q14	2C "	27	25	2	"	"	"	"	"
Q15	3C "	8	28	30	"	"	"	"	"
Q16	4C "	26	7	4	"	"	"	"	"
Q17	5C "	6	5	24	"	"	"	"	"
Q18	6C "	29	13	9	"	"	"	"	"
Q19	7C "	32	12	31	"	"	"	"	"
Q20	8C "	14	33	11	"	"	"	"	"
Q21	9C "	34	15	37	"	"	"	"	"
Q22	10C "	35	39	19	"	"	"	"	"
Q23	11C "	16	36	17	"	"	"	"	"
Q24	12C "	20	40	38	"	"	"	"	"
R1	1C Propagation delay time (t_{PHL}), C_L = 50 pF	22	21	23	"	3	19	"	"
R2	2C "	25	27	2	"	"	"	"	"
R3	3C "	28	8	30	"	"	"	"	"
R4	4C "	7	26	4	"	"	"	"	"
R5	5C "	5	6	24	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R6	6C Propagation delay time (t_{PHL}), C_L = 50 pF	13	29	9	Open	3	19	ns	17,18
R7	7C "	12	32	31	"	"	"	"	"
R8	8C "	33	14	11	"	"	"	"	"
R9	9C "	15	34	37	"	"	"	"	"
R10	10C "	39	35	19	"	"	"	"	"
R11	11C "	36	16	17	"	"	"	"	"
R12	12C "	40	20	38	"	"	"	"	"
R13	1C Propagation delay time (t_{PHL}), C_L = 150 pF	22	21	23	"	4	29	"	"
R14	2C "	25	27	2	"	"	"	"	"
R15	3C "	28	8	30	"	"	"	"	"
R16	4C "	7	26	4	"	"	"	"	"
R17	5C "	5	6	24	"	"	"	"	"
R18	6C "	13	29	9	"	"	"	"	"
R19	7C "	12	32	31	"	"	"	"	"
R20	8C "	33	14	11	"	"	"	"	"
R21	9C "	15	34	37	"	"	"	"	"
R22	10C "	39	35	19	"	"	"	"	"
R23	11C "	36	16	17	"	"	"	"	"
R24	12C "	40	20	38	"	"	"	"	"

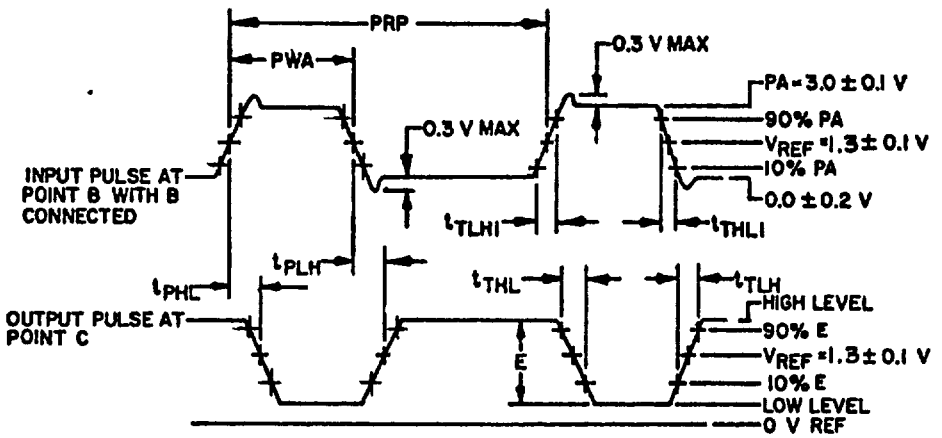
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

*FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 6 \pm 1$ ns.
   PWA = 500 ±50 ns.
   PRP = 1000 ±100 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.
3. The lead length of electrical conductors between resistor R1 and input pins; items Y01-1 and Y01-2 to input and output pins; and between load circuit and output pin shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

*FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk (low-level and high-level) test.

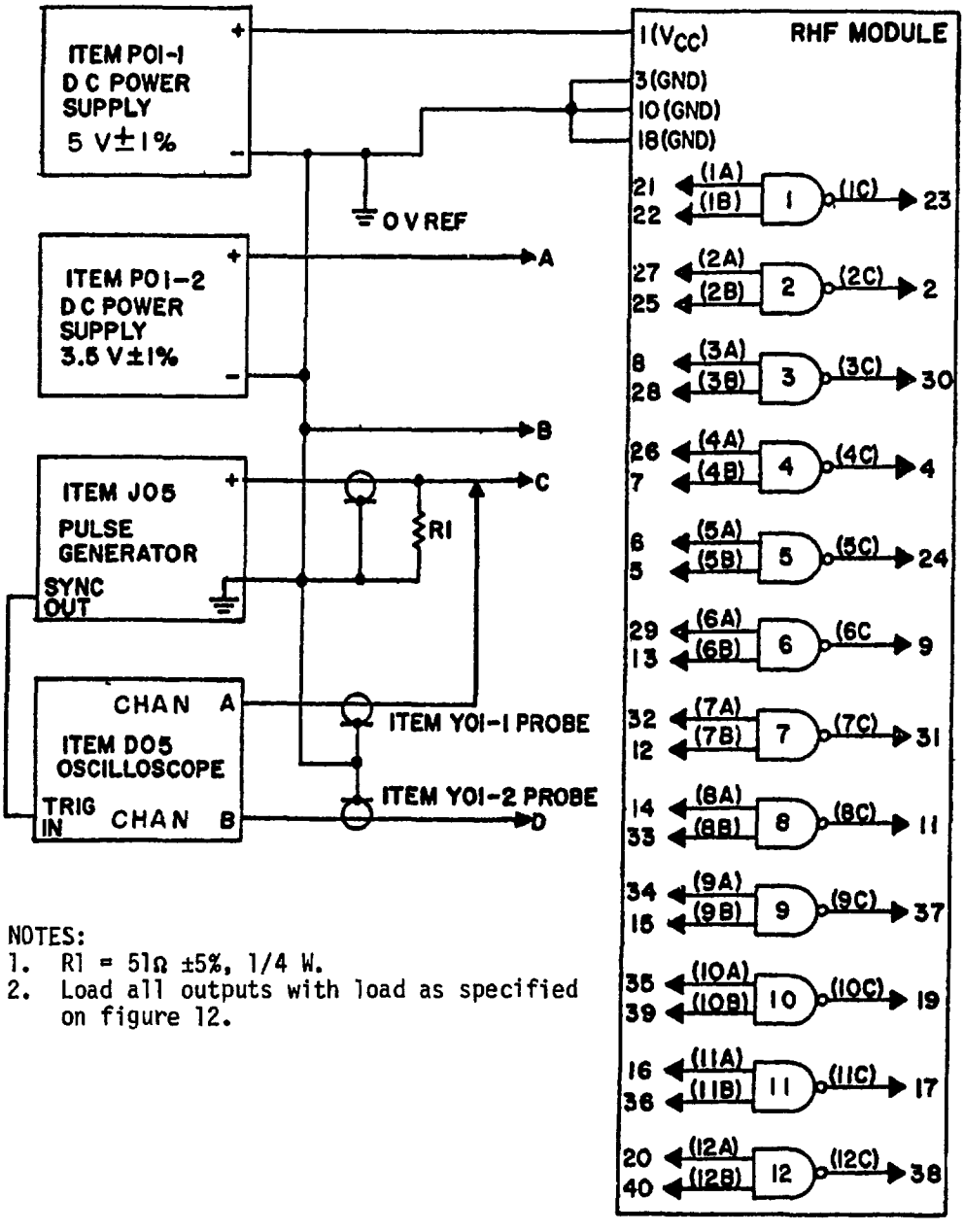
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1C Output crosstalk, low level	5,7,12,13,15,21,22, 25,28,33,36,39,40	Open	6,8,14,16,20,26,27, 29,32,34,35	23	0	500	mV	19,20
S2	2C "	5,7,12,13,15,22,25, 27,28,33,36,39,40	"	6,8,14,16,20,21,26, 29,32,34,35	2	"	"	"	"
S3	3C "	5,7,8,12,13,15,22, 25,28,33,36,39,40	"	6,14,16,20,21,26,27, 29,32,34,35	30	"	"	"	"
S4	4C "	5,7,12,13,15,22,25, 26,28,33,36,39,40	"	6,8,14,16,20,21,27, 29,32,34,35	4	"	"	"	"
S5	5C "	5,6,7,12,13,15,22, 25,28,33,36,39,40	"	8,14,16,20,21,26,27, 29,32,34,35	24	"	"	"	"
S6	6C "	5,7,12,13,15,22,25, 28,29,33,36,39,40	"	6,8,14,16,20,21,26, 27,32,34,35	9	"	"	"	"
S7	7C "	5,7,12,13,15,22,25, 28,32,33,36,39,40	"	6,8,14,16,20,21,26, 27,29,34,35	31	"	"	"	"
S8	8C "	5,7,12,13,14,15,22, 25,28,33,36,39,40	"	6,8,16,20,21,26,27, 29,32,34,35	11	"	"	"	"
S9	9C "	5,7,12,13,15,22,25, 28,33,34,36,39,40	"	6,8,14,16,20,21,26, 27,29,32,35	37	"	"	"	"
S10	10C "	5,7,12,13,15,22,25, 28,33,35,36,39,40	"	6,8,14,16,20,21,26, 27,29,32,34	19	"	"	"	"
S11	11C "	5,7,12,13,15,16,22, 25,28,33,36,39,40	"	6,8,14,20,21,26,27, 29,32,34,35	17	"	"	"	"
S12	12C "	5,7,12,13,15,20,22, 25,28,33,36,39,40	"	6,8,14,16,21,26,27, 29,32,34,35	38	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XVI. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
T1	1C Output crosstalk, high level	5,7,12,13,15,25,28, 33,36,39,40	21,22	6,8,14,16,20,26,27, 29,32,34,35	23	0	400	mV	19,20
T2	2C "	5,7,12,13,15,22,28, 33,36,39,40	25,27	6,8,14,16,20,21,26, 29,32,34,35	2	"	"	"	"
T3	3C "	5,7,12,13,15,22,25, 33,36,39,40	8,28	6,14,16,20,21,26,27, 29,32,34,35	30	"	"	"	"
T4	4C "	5,12,13,15,22,25, 28,33,36,39,40	7,26	6,8,14,16,20,21,27, 29,32,34,35	4	"	"	"	"
T5	5C "	7,12,13,15,22,25, 28,33,36,39,40	5,6	8,14,16,20,21,26,27, 29,32,34,35	24	"	"	"	"
T6	6C "	5,7,12,15,22,25,28, 33,36,39,40	13,29	6,8,14,16,20,21,26, 27,32,34,35	9	"	"	"	"
T7	7C "	5,7,13,15,22,25,28, 33,36,39,40	12,32	6,8,14,16,20,21,26, 27,29,34,35	31	"	"	"	"
T8	8C "	5,7,12,13,15,22,25, 28,36,39,40	14,33	6,8,16,20,21,26,27, 29,32,34,35	11	"	"	"	"
T9	9C "	5,7,12,13,22,25,28, 33,36,39,40	15,34	6,8,14,16,20,21,26, 27,29,32,35	37	"	"	"	"
T10	10C "	5,7,12,13,15,22,25, 28,33,36,40	35,39	6,8,14,16,20,21,26, 27,29,32,34	19	"	"	"	"
T11	11C "	5,7,12,13,15,22,25, 28,33,39,40	16,36	6,8,14,20,21,26,27, 29,32,34,35	17	"	"	"	"
T12	12C "	5,7,12,13,15,22,25, 28,33,36,39	20,40	6,8,14,16,21,26,27, 29,32,34,35	38	"	"	"	"

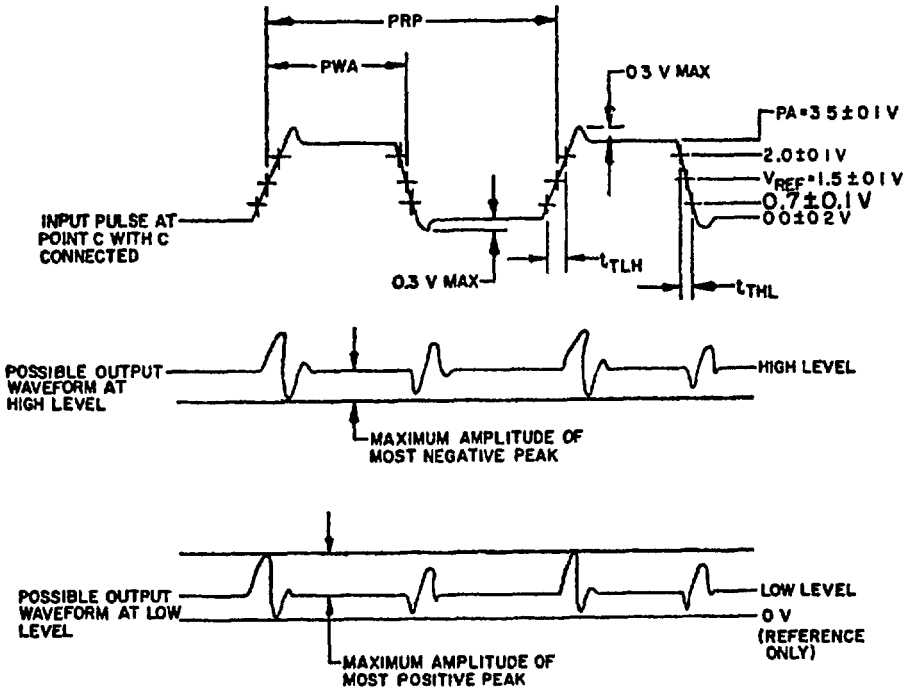
Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/4 W.
2. Load all outputs with load as specified on figure 12.

FIGURE 19. Output crosstalk (low-level and high-level) test circuit for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:

1. Input pulse parameters: For each test condition with the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making each measurement.

   $t_{THL} = t_{TLH} = 6 \pm 2$ ns (measured at the 0.7 V and 2.0 V points of the input pulse).
   PWA = 500 ±50 ns.
   PRP = 1,000 ±100 ns.

2. The lead length of electrical conductors between resistor R1 and input pins; items Y01-1 and Y01-2 to input and output pins; load circuits to output pins; and ground leads of items Y01-1 and Y01-2 to test fixture and module ground pins shall be less than 1 inch (25.4 mm).

FIGURE 20. Output crosstalk (low-level and high-level) test circuit waveforms for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XVII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
U	Power supply transient amplitude	0	400	mV	21,22

TABLE XVIII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
V	Power supply filter capacitance	Open	1	1.6	3.0	µF	23

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XIX. Input capacitance test.

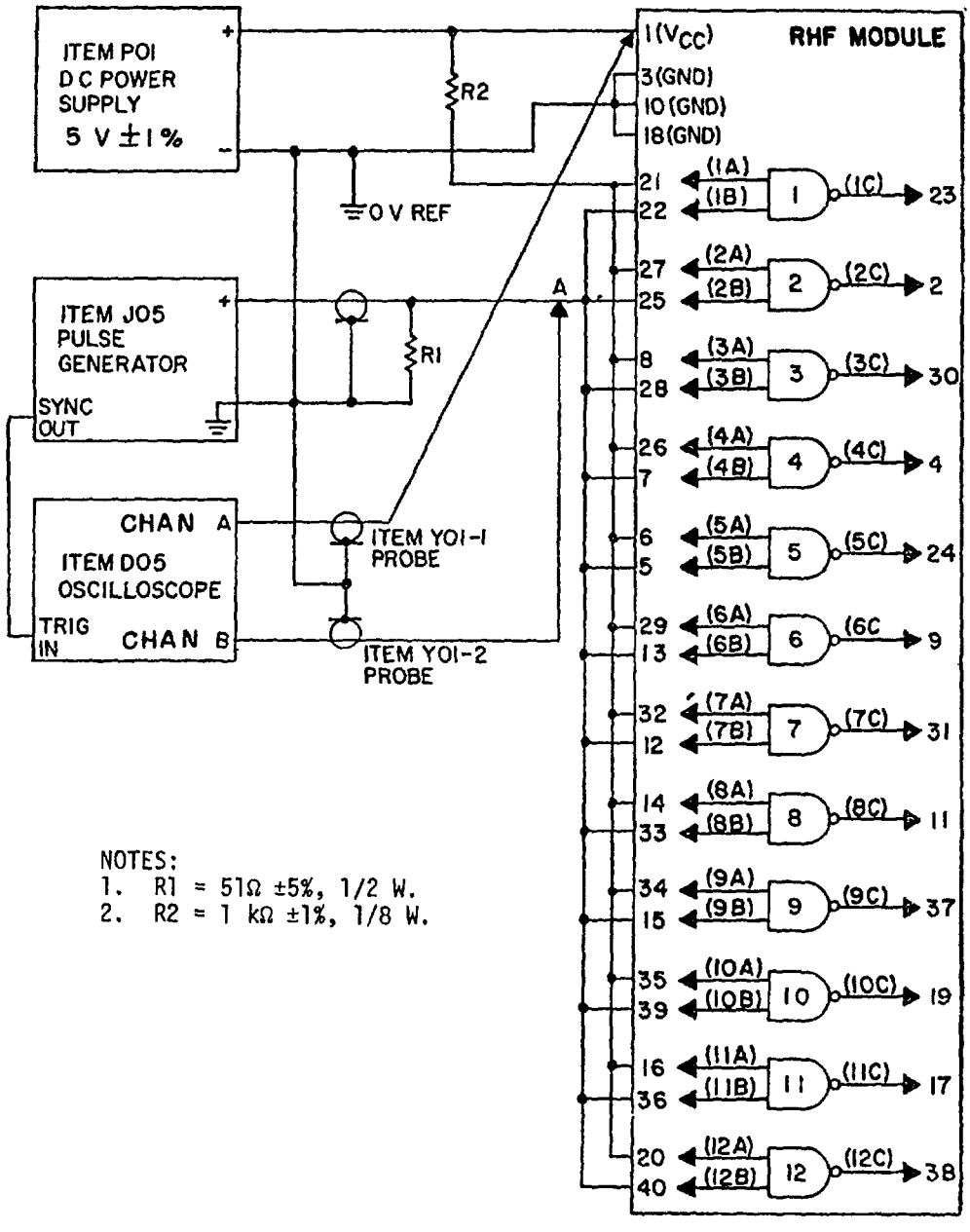
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W1	1A Input capacitance	21	Open	5	15	pF	23
W2	1B "	22	"	"	"	"	"
W3	2A "	27	"	"	"	"	"
W4	2B "	25	"	"	"	"	"
W5	3A "	8	"	"	"	"	"
W6	3B "	28	"	"	"	"	"
W7	4A "	26	"	"	"	"	"
W8	4B "	7	"	"	"	"	"
W9	5A "	6	"	"	"	"	"
W10	5B "	5	"	"	"	"	"
W11	6A "	29	"	"	"	"	"
W12	6B "	13	"	"	"	"	"
W13	7A "	32	"	"	"	"	"
W14	7B "	12	"	"	"	"	"
W15	8A "	14	"	"	"	"	"
W16	8B "	33	"	"	"	"	"
W17	9A "	34	"	"	"	"	"
W18	9B "	15	"	"	"	"	"
W19	10A "	35	"	"	"	"	"
W20	10B "	39	"	"	"	"	"

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.

TABLE XIX. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W21	11A Input capacitance	16	Open	5	15	pF	23
W22	11B "	36	"	"	"	"	"
W23	12A "	20	"	"	"	"	"
W24	12B "	40	"	"	"	"	"

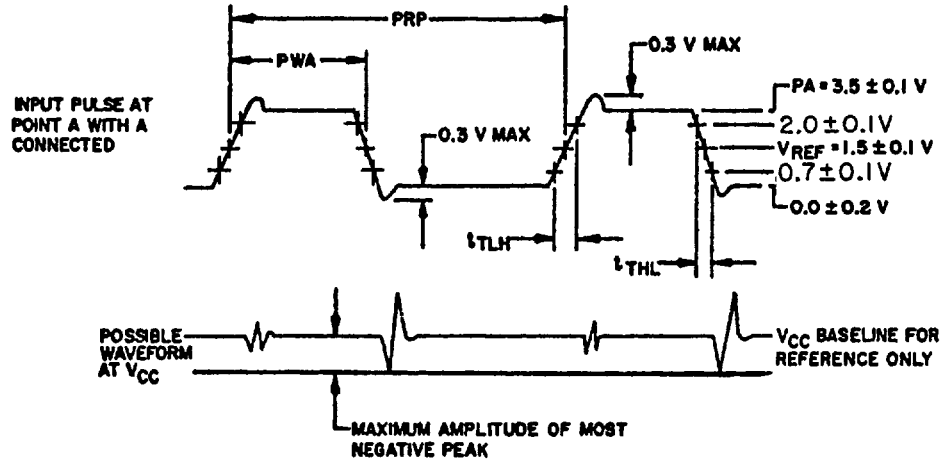
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. R2 = 1 kΩ ±1%, 1/8 W.

FIGURE 21. Power supply transient amplitude test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.

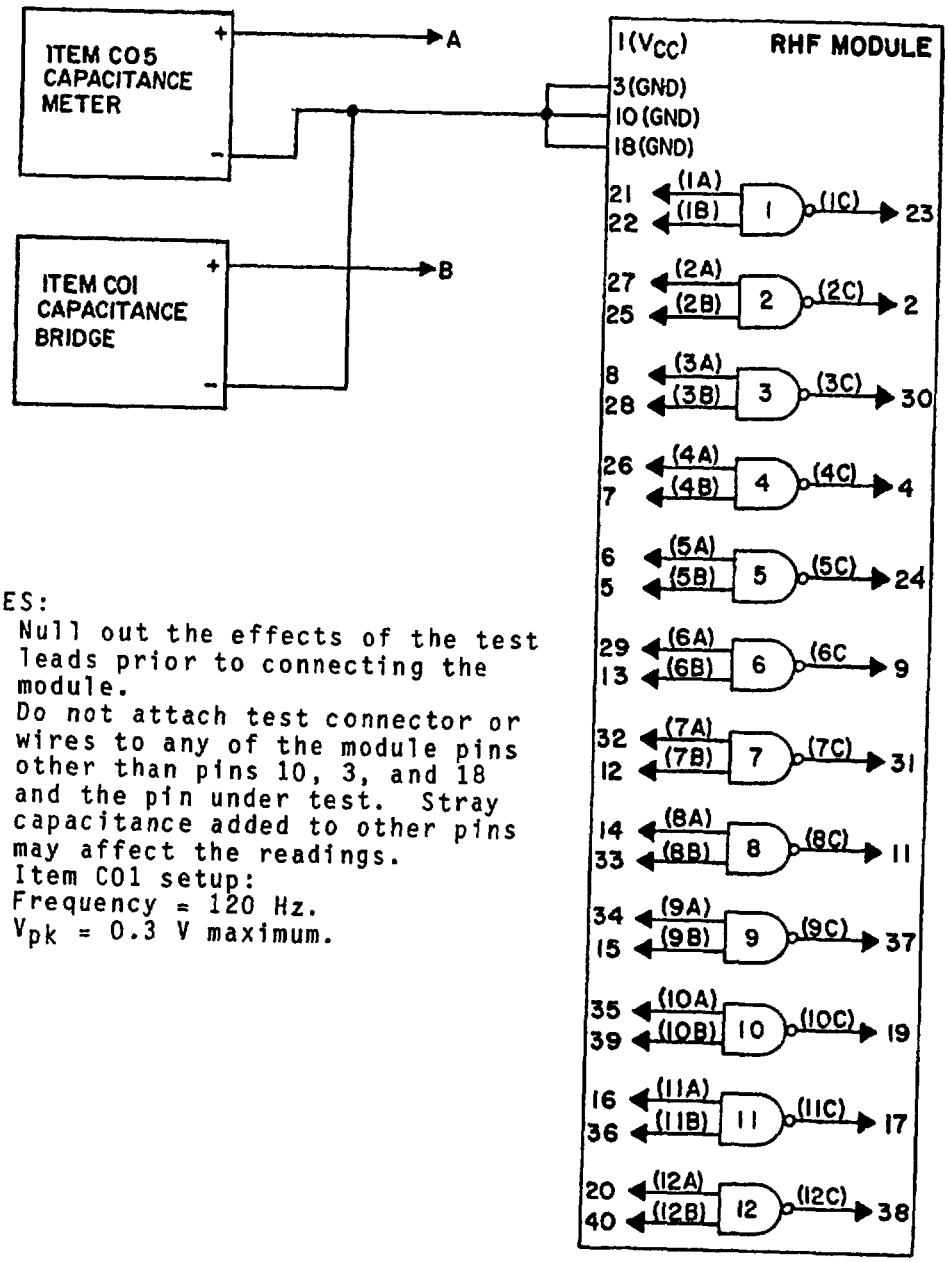


NOTE: Input pulse parameters: With the module connected to the pulse generator, adjust the pulse generator for the required pulse amplitude, pulse low level voltage, and transition times. Verify the input pulse parameters at each input pin before making the measurement.

PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.
$t_{TLH} = t_{THL}$ = 3.5 ±1.5 ns verified at each input pin.

FIGURE 22. Power supply transient amplitude test circuit waveforms for table XVII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 10, 3, and 18 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item CO1 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

FIGURE 23. Power supply filter and input capacitance test circuit for tables XVIII and XIX.

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:29Z
Check the source to verify that this is the current version before use.

STRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in ubmitting comments and suggestions for improvements. All users of military standardization documents are invited to provide ggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and ailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was o rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the roblems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an cknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being onsidered.

OTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of pecification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization o waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

Naval Electronic Systems Command
Washington, DC 20363

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

FFICIAL BUSINESS
ENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

COMMANDER
NAVAL ELECTRONIC SYSTEMS COMMAND
DEFENSE STANDARDIZATION PROGRAM BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, DC 20363
ATTN: ELEX 8111

Source: https://assist.dla.mil -- Downloaded: 20260-20T17:29Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1 DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-28787/297A(EC)	Modules, Standard Electronic Gate, Nand, Digital, Key Code RHF

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5 PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6 REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:29Z
Check the source to verify that this is the current version before use.